From Person-Specific Dynamics to Population Inference: A Meta-Analytic Structural Equation Modeling Framework for Multilevel Dynamic Models

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Abstract

Intensive longitudinal studies increasingly seek population-level conclusions about within-person dynamics while preserving person-specific estimation. Extending prior idiographic two-stage meta-analytic work to multivariate dynamic systems, we introduce MetaVAR, a two-stage meta-analytic structural equation modeling (MASEM) framework for multilevel dynamic models. In Stage 1, person-specific vector autoregressive (VAR) models are fit idiographically to obtain a parameter vector and its sampling covariance matrix for each individual. In Stage 2, these person-level parameter vectors are synthesized using a multivariate random-effects meta-analytic SEM, allowing full propagation of first-stage uncertainty and system-level inference on the joint dynamic parameter system. This formulation yields population summaries of typical dynamics, between-person heterogeneity, and cross-parameter co-variation, and it naturally supports constraints and extensions involving covariates or distal outcomes. We evaluate MetaVAR in a Monte Carlo simulation against a Bayesian multilevel VAR estimator and a naïve fit-many-then-summarize approach. Results show that MetaVAR is especially advantageous for recovering between-person heterogeneity and for producing better-calibrated interval estimates for heterogeneity parameters, while one-step hierarchical modeling remains highly competitive for population-average effects. We then illustrate the approach using ecological momentary assessment data on negative and positive affect, showing how MetaVAR yields interpretable population-average dynamics together with scientifically meaningful heterogeneity in affective set points, temporal persistence, cross-lagged coupling, and innovation structure. MetaVAR provides a principled middle ground between one-step hierarchical estimation and uncorrected two-stage aggregation for dynamic psychological processes.

Keywords: vector autoregressive model, multilevel model, random-effects meta-analysis

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Intensive longitudinal designs—such as ecological momentary assessment (EMA) and other experience-sampling, daily diary, ambulatory assessment, and measurement-burst protocols—have made it increasingly feasible to collect dense within-person data in everyday contexts (Bolger & Laurenceau, 2013; Bolger et al., 2003; Csikszentmihalyi & Larson, 1987; Nesselroade, 1991; Shiffman et al., 2008; Sliwinski, 2008; Trull & Ebner-Priemer, 2013). These designs make it increasingly feasible to study psychological processes as dynamic systems, where within-person trajectories evolve over time and individuals differ meaningfully in the parameters that govern those trajectories (Asparouhov et al., 2018; Bringmann et al., 2013, 2016; Fisher et al., 2011; Hamaker et al., 2018; Rowland & Wenzel, 2020). This shift has reinforced a long-standing methodological point: inferences about within-person dynamics cannot be taken for granted from between-person variation, and scientific questions about regulation, coupling, and stability often require models that respect the distinction between idiographic (person-specific) and nomothetic (population-level) structure (Curran & Bauer, 2011; Fisher et al., 2018; Gates et al., 2023; Hamaker et al., 2005; Molenaar, 2004; Wang et al., 2026). At the same time, applied researchers rarely want only a collection of person-specific estimates; they also want principled summaries of what is typical, how much people differ, how dynamic features co-vary across people, and how these dynamic "signatures" relate to covariates and distal outcomes (Beltz et al., 2016; Li et al., 2022). Yet existing workflows often force a trade-off between (a) one-step hierarchical models with partial pooling—including multilevel and Bayesian multilevel formulations—that can be computationally demanding and shrink person-specific dynamics and (b) "fit-many-then-summarize" approaches that are fast but typically ignore first-stage uncertainty and the multivariate dependence among dynamic parameters—limiting system-level inference and rapid model iteration (Moeyaert et al., 2016; Van den Noortgate & Onghena, 2008).

To meet these applied goals, we introduce MetaVAR, a meta-analytic structural equation modeling (MASEM; Cheung, 2008, 2013, 2015) framework for vector autoregressive (VAR; Hamilton, 1994; Lütkepohl, 2005) systems. MetaVAR[1] is a two-stage approach that (a) estimates person-specific VAR dynamics and (b) performs *system-level* synthesis of the resulting parameter vectors using multivariate random-effects MASEM with full propagation of first-stage uncertainty. By carrying forward each individual's full first-stage sampling covariance, MetaVAR yields population summaries of typical dynamics, heterogeneity, and cross-parameter covariation. When desired, the same synthesis stage can be extended to relate dynamic signatures to covariates and distal outcomes while retaining Stage 1 error propagation.

Against this backdrop, the key methodological challenge is to draw population-level conclusions about within-person dynamics without sacrificing inferential validity or practical feasibility. Current workflows tend to fall at two ends of a spectrum: one-step hierarchical models with partial pooling, which can be computationally demanding and induce shrinkage in person-specific dynamics, and fully unpooled fit-many approaches that scale well but often treat first-stage estimates as error-free. In the next section, we review these dominant strategies and argue that both leave a persistent gap: a principled two-stage workflow that (a) performs system-level synthesis with full propagation of Stage 1 uncertainty and (b) supports theory-driven constraints, with optional extensions for covariate moderation and distal outcomes. This gap motivates MetaVAR, which we

[1] We use the label MetaVAR because, in this manuscript, the Stage 1 idiographic model is a VAR fit separately to each individual. Importantly, the proposed two-stage framework is not tied to VAR dynamics. Stage 2 requires only that each person contribute (a) a vector of estimated model features, $\mathbf{y}_i$, and (b) an accompanying sampling covariance matrix, $\mathbf{V}_i^2$, so that first-stage uncertainty is propagated into the population synthesis. Thus, MetaVAR may be viewed as a special case of a more general meta-analytic dynamical-systems template in which person-specific parameters (or derived functionals thereof) from a wide class of dynamical models—including state-space, continuous-time, and nonlinear formulations—are synthesized using the same multivariate mixed-effects SEM machinery.

evaluate in a Monte Carlo simulation, illustrate with an empirical example, and then discuss in terms of practical guidance, limitations, and future extensions.

Two Dominant Strategies and a Persistent Gap

Two broad strategies dominate current practice for drawing population-level conclusions from intensive longitudinal multivariate time series. The first is one-step hierarchical (partially pooled) modeling—including multilevel and Bayesian multilevel formulations such as dynamic structural equation modeling (DSEM) and Bayesian multilevel VAR (mlVAR)—in which within-person dynamics and between-person variation are estimated simultaneously within a single hierarchical model (Asparouhov et al., 2018; Hamaker et al., 2018; Li et al., 2022). This approach is attractive in the behavioral sciences because the per-person series length is often limited, making it beneficial to "borrow strength" across individuals while still allowing random effects in dynamic parameters (Li et al., 2022). At the same time, the resulting estimation problem can become computationally demanding as the number of variables and random effects grows, and practical implementations often require simplifying assumptions about the random-effects structure or otherwise face feasibility and convergence constraints in high-dimensional settings (Asparouhov et al., 2018; Epskamp et al., 2018; Li et al., 2022). Moreover, because hierarchical estimators partially pool information, person-specific dynamics are shrunken toward the population mean—a feature that can improve stability and out-of-sample performance (Gelman & Hill, 2006). At the same time, shrinkage means that the resulting person-level estimates are biased toward the center and may be less suitable as direct measures of individual differences or for detecting unusual individuals (Efron & Morris, 1975).

At the other extreme are *fully unpooled* workflows that fit an idiographic model per person and then summarize the resulting coefficients informally (e.g., by averaging). This "fit-many-then-summarize" approach is modular and naturally parallelizable, but naïve summaries ignore that individuals' estimates differ in precision as a function of series length, missingness, and measurement quality, and simple averaging fails to weight estimates by their uncertainty (Hedges & Olkin, 1985; Pastor & Lazowski, 2017). In intensive time-series settings, dependence and model misspecification (e.g., unmodeled trends) can also induce artifactual residual autocorrelation and larger residuals, further affecting precision and inference if not modeled (Baek et al., 2023; Shadish et al., 2008). In multivariate dynamic systems, a second limitation is that piecemeal estimation (e.g., collections of univariate models) need not place all parameters in a single joint model, making it difficult to retain the within-person sampling covariance among dynamic parameters that is required for principled system-level synthesis and inference (Epskamp et al., 2018; Moeyaert et al., 2016). Accordingly, valid two-stage synthesis requires carrying forward each individual's first-stage sampling variance (and, in multivariate settings, the full sampling covariance) into the second stage rather than treating person-level estimates as error-free (Van den Noortgate & Onghena, 2008).

A natural middle ground is *two-stage* inference: estimate person-specific dynamic models first, then integrate those results to make population statements about within-person processes. However, two-stage workflows only deliver valid population inference when the second stage correctly accounts for first-stage estimation error and, in multivariate settings, the dependence among estimated parameters. The present work develops a two-stage framework that fills this gap for dynamic systems by treating the synthesis stage as a multivariate meta-analytic structural equation model.

Related Two-Stage Literatures: Idiographic Meta-Analysis and Error-Corrected Pooling

MetaVAR builds on, and sits alongside, two closely related two-stage literatures. First, Lee and Gates (2023) advocate a *two-stage idiographic meta-analytic* perspective for observational time series, emphasizing the importance of estimating within-person models idiographically and then using random-effects meta-analysis to characterize population distributions of within-person effects. This perspective foregrounds heterogeneity as a scientific target. MetaVAR adopts that stance and shares its two-stage spirit, but broadens the target of synthesis. Whereas Lee and Gates (2023) articulate the case for aggregating idiographic time-series results via random-effects meta-analysis and illustrate path-specific population inference for selected network relations, MetaVAR synthesizes the full person-specific dynamic parameter vector by carrying forward the full Stage-1 sampling covariance matrix. This extension preserves dependence among within-person effects, supports coherent inference under cross-parameter constraints, and enables valid uncertainty propagation to functions of the dynamics (e.g., stability indices, impulse responses, and network summaries). It also addresses several next-step issues they identify, including when one-stage and two-stage approaches may yield comparable inferences, how interval estimation behaves in two-stage settings, and how to move beyond one-coefficient-at-a-time synthesis.

Second, Zhang et al. (2025) develop an *error-corrected two-stage* approach for integrating individual-level dynamic factor analysis (DFA) parameters, showing that ignoring first-stage estimation error can bias second-stage random-effects estimation and demonstrating an efficient correction. MetaVAR generalizes this error-propagation logic to VAR systems and uses an SEM-based synthesis stage to support *system-level* inference, including joint hypotheses, cross-parameter constraints, and second-stage models defined on the full parameter vector.

Taken together, these strands motivate a two-stage workflow that is both conceptually idiographic and inferentially principled; MetaVAR operationalizes this combination for VAR models by casting the second stage as multivariate meta-analytic structural equation modeling.

The move from path-specific synthesis to multivariate system-level synthesis is not merely a technical generalization. In dynamic systems, substantive questions often concern parameter configurations rather than isolated coefficients—for example, whether stronger autoregressive persistence co-occurs with stronger cross-lagged coupling, or whether summary functionals of the system vary systematically across persons. A multivariate second stage is therefore needed when the scientific target is the dynamic system as a whole rather than any single path in isolation.

MetaVAR: Dynamic MASEM for System-Level Pooling With Full Error Propagation

MetaVAR is a two-stage framework that treats multilevel dynamic modeling as a meta-analytic problem in which *persons play the role of studies*. In Stage 1, each individual $i = 1, \ldots, N$ is fit with an idiographic dynamic model (e.g., VAR, latent-variable VAR, or state-space model), yielding a vector of estimated dynamic parameters and an accompanying sampling covariance matrix. In Stage 2, these person-level parameter vectors are synthesized using a multivariate mixed-effects meta-analytic model expressed as a structural equation model. This design yields *system-level pooling with full error propagation*: MetaVAR preserves the full within-person sampling covariance among estimated parameters from Stage 1 and simultaneously estimates a between-person random-effects covariance structure for the entire parameter system. The perspective connects directly to multivariate random-effects meta-analysis (Berkey et al., 1998; van Houwelingen et al., 1993, 2002) and MASEM (Cheung, 2008, 2013, 2015) and directly targets questions central to intensive longitudinal research: What is the typical within-person dynamic system? How heterogeneous is it? Which dynamic features co-occur across people? Which person-level covariates predict dynamic regimes? How do dynamics relate to clinically meaningful distal outcomes?

Stage 1 Outputs as Multivariate Effect Sizes

Let $\mathbf{y}_i \in \mathbb{R}^p$ denote the vector of person-i parameter estimates returned by the Stage 1 model, treated as a multivariate set of meta-analytic effect sizes, and let $\mathbf{V}_i^2 \in \mathbb{R}^{p \times p}$ denote the corresponding Stage 1 sampling covariance matrix. The contents of $\mathbf{y}_i$ are deliberately flexible: depending on the Stage 1 model, $\mathbf{y}_i$ may include autoregressive and cross-lag coefficients, innovation variances and covariances, set point (mean) parameters, measurement-error variances, or other person-specific system features. The only requirement for MetaVAR is that Stage 1 yields $\mathbf{V}_i^2$ (or a good approximation), so that Stage 2 can propagate Stage 1 uncertainty rather than treating the person-level estimates as error-free.

In this manuscript, Stage 1 fits a person-specific first-order VAR model. Let $\mathbf{s}_{i,t} \in \mathbb{R}^k$ denote the observed multivariate time series for person i at occasion $t = 1, \ldots, T$. The dynamic model follows a centered VAR model (Asparouhov & Muthén, 2018; Hamaker & Grasman, 2015),

$$\mathbf{s}_{i,t} = \boldsymbol{\mu}_i + \mathbf{A}_i \left(\mathbf{s}_{i,t-1} - \boldsymbol{\mu}_i\right) + \boldsymbol{\zeta}_{i,t}, \quad \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Xi}_i\right), \tag{1}$$

where $\boldsymbol{\mu}_i$ is the person-specific set point (i.e., the equilibrium or long-run mean level of the process), $\mathbf{A}_i$ collects autoregressive and cross-lagged effects that govern the evolution of deviations from this set point, and $\boldsymbol{\Xi}_i$ is the innovation covariance. Under stability/stationarity (e.g., all eigenvalues of $\mathbf{A}_i$ lie inside the unit circle) and mean-zero innovations, the process fluctuates around $\boldsymbol{\mu}_i$ such that $\mathbb{E}(\mathbf{s}_{i,t}) = \boldsymbol{\mu}_i$.

To connect Stage 1 to Stage 2, we stack the person-specific parameters into a single vector $\boldsymbol{\theta}_i \in \mathbb{R}^p$ and take $\mathbf{y}_i = \hat{\boldsymbol{\theta}}_i$. A convenient choice for the present model is

$$\boldsymbol{\theta}_i = \left(\boldsymbol{\mu}_i', \operatorname{vec}(\mathbf{A}_i)', \operatorname{vech}(\boldsymbol{\Xi}_i)'\right)', \tag{2}$$

where $(\cdot)'$ represents matrix transpose, $\operatorname{vec}(\cdot)$ stacks a matrix by columns and $\operatorname{vech}(\cdot)$ stacks the lower triangle (including the diagonal) of a symmetric matrix. The associated Stage 1 sampling covariance matrix $\mathbf{V}_i^2 = \widehat{\mathbb{V}}\left[\hat{\boldsymbol{\theta}}_i\right]$ can be obtained from the observed information matrix, a sandwich estimator, or a parametric bootstrap, and is passed to Stage 2 to propagate first-stage estimation uncertainty. If some components are fixed or omitted in a given application, they are simply excluded from $\boldsymbol{\theta}_i$ and the dimension p is adjusted accordingly.

Stage 2 as Multivariate Mixed-Effects Meta-Analysis in SEM Form

A general mixed-effects meta-analytic SEM can be written as (Cheung, 2008, 2013)

$$\mathbf{y}_i = \boldsymbol{\alpha}_i + \boldsymbol{\beta}\mathbf{y}_i + \boldsymbol{\Gamma}\mathbf{x}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \mathbf{V}_i^2\right), \tag{3}$$

with person-specific "true" effect vectors

$$\boldsymbol{\alpha}_i = \boldsymbol{\alpha} + \boldsymbol{\upsilon}_i, \quad \boldsymbol{\upsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\tau}^2\right). \tag{4}$$

Here, $\boldsymbol{\alpha}$ is the grand-mean vector of the dynamic-parameter system, $\boldsymbol{\tau}^2$ is the between-person heterogeneity covariance matrix, $\mathbf{x}_i$ collects person-level covariates (e.g., traits, clinical status, context), and $\boldsymbol{\beta}$ and $\boldsymbol{\Gamma}$ allow researchers to specify structured relations and meta-regressions (moderation) defined *on the parameter system* itself.

Assuming that $\mathbf{x}_i$ is non-stochastic and that $\mathbf{I} - \boldsymbol{\beta}$ is nonsingular, the model can be written in reduced form (Bollen, 1989) as

$$\mathbf{y}_i = (\mathbf{I} - \boldsymbol{\beta})^{-1} (\boldsymbol{\alpha}_i + \boldsymbol{\Gamma}\mathbf{x}_i + \boldsymbol{\varepsilon}_i). \tag{5}$$

Therefore, conditional on $\mathbf{x}_i$, the marginal model implied mean vector and covariance matrix[2] are

$$\begin{aligned} \mathbb{E}[\mathbf{y}_i \mid \mathbf{x}_i] &= (\mathbf{I} - \boldsymbol{\beta})^{-1} (\boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{x}_i), \\ \mathbb{V}[\mathbf{y}_i \mid \mathbf{x}_i] &= (\mathbf{I} - \boldsymbol{\beta})^{-1} \left(\boldsymbol{\tau}^2 + \mathbf{V}_i^2\right) (\mathbf{I} - \boldsymbol{\beta})^{-1\prime}. \end{aligned} \tag{6}$$

[2] In SEM software that supports *definition variables*, elements of the known, person-specific sampling covariance matrix $\mathbf{V}_i^2$ can be supplied as case-level data so that the model-implied covariance is recomputed for each individual (Mehta & Neale, 2005). This capability is central in meta-analytic applications because $\mathbf{V}_i^2$ is treated as a known input (and can be fixed as such), whereas common mixed-effects implementations (e.g., `lmer`) do not directly support fixing a case-specific sampling covariance matrix in the likelihood (Declercq et al., 2020). In `OpenMx`, this is implemented by evaluating each individual's multivariate normal log-likelihood using that individual's model-implied mean vector and covariance matrix, which may depend on definition variables (FIML; Arbuckle, 1996; Enders, 2010; Neale et al., 2015). The `metaSEM` package leverages this capability via `OpenMx` (Cheung, 2015); the `metaDyn` implementation of MetaVAR uses the same definition-variable strategy to incorporate $\mathbf{V}_i^2$ and propagate Stage 1 uncertainty through Stage 2.

In the special case $\boldsymbol{\beta} = \mathbf{0}$, the model reduces to a multivariate mixed-effects meta-analysis (i.e., multivariate meta-regression when $\boldsymbol{\Gamma} \neq \mathbf{0}$), with $\mathbb{E}[\mathbf{y}_i \mid \mathbf{x}_i] = \boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{x}_i$ and $\mathbb{V}[\mathbf{y}_i \mid \mathbf{x}_i] = \boldsymbol{\tau}^2 + \mathbf{V}_i^2$. When no covariates are included and $\boldsymbol{\beta} = \mathbf{0}$, the model further reduces to a multivariate random-effects meta-analysis with $\mathbb{E}[\mathbf{y}_i] = \boldsymbol{\alpha}$ and $\mathbb{V}[\mathbf{y}_i] = \boldsymbol{\tau}^2 + \mathbf{V}_i^2$. Thus, the model decomposes observed dispersion in $\mathbf{y}_i$ into (a) within-person estimation uncertainty carried by $\mathbf{V}_i^2$ and (b) genuine between-person heterogeneity carried by $\boldsymbol{\tau}^2$ (Pastor & Lazowski, 2017).

This system-level viewpoint is the key departure from edge-by-edge pooling. MetaVAR treats the full parameter vector as the unit of synthesis, which (a) retains the entire within-person sampling covariance among dynamic parameters and (b) estimates a coherent between-person covariance structure for the dynamic system. As a descriptive summary, heterogeneity can be reported using multivariate generalizations of $\mathbf{I}^2$ (Higgins & Thompson, 2002; Jackson et al., 2012; Viechtbauer, 2007), for example

$$\mathbf{I}^2 = p^{-1} \left\{ \mathrm{tr} \left[\boldsymbol{\tau}^2 \left(\boldsymbol{\tau}^2 + \bar{\mathbf{V}}^2 \right)^{-1} \right] \right\}. \tag{7}$$

Here, $\mathrm{tr}(\cdot)$ denotes the matrix trace (the sum of diagonal elements), and $\bar{\mathbf{V}}^2$ is the harmonic (precision-weighted) mean of the Stage 1 sampling covariance matrices,

$$\bar{\mathbf{V}}^2 = \left[N^{-1} \sum_{i=1}^{N} \left(\mathbf{V}_i^2 \right)^{-1} \right]^{-1}, \tag{8}$$

which gives greater weight to individuals with more precise Stage 1 estimates (smaller sampling variances). Values near zero indicate that most observed dispersion in person-level parameters reflects estimation uncertainty, whereas values near one indicate substantial true heterogeneity in the underlying dynamic system.

Model-Level Hypotheses and Constraints

Because Stage 2 is an SEM defined on the full parameter vector, MetaVAR is naturally aligned with *model-level* hypotheses—that is, hypotheses that constrain or compare *multiple* dynamic parameters jointly rather than treating each coefficient as an isolated edge. This matters for applied research because many substantive predictions about regulation are inherently multivariate: for example, whether cross-lag effects are asymmetric ($A_{1,2} \neq A_{2,1}$), whether a subset of couplings is negligible relative to others (block-zero constraints), whether inertia parameters are comparable across constructs (equality constraints), or whether a theoretically motivated contrast among parameters differs from zero (e.g., a "self–other" coupling contrast).

In MetaVAR, these hypotheses are expressed directly through standard SEM constraint machinery applied to $\boldsymbol{\alpha}$, $\boldsymbol{\tau}^2$, $\boldsymbol{\beta}$, and $\boldsymbol{\Gamma}$. This yields several practical benefits. First, inference is *system-aware*: constraints are evaluated while retaining the within-person sampling covariance $\mathbf{V}_i^2$ and the between-person heterogeneity structure $\boldsymbol{\tau}^2$, so uncertainty is not understated by treating estimated parameters as independent or error-free. Second, constraints can be placed not only on individual coefficients but also on *functions* of the dynamic system that are often the actual scientific target (e.g., stability-related summaries, coupling differences, or composite indices that pool information across parameters). Third, competing theory-level models can be compared transparently using standard SEM tools (e.g., likelihood-based model comparison for nested specifications or information criteria for non-nested structures), enabling researchers to formalize "which dynamic story fits best" at the level of the full parameter system.

In this manuscript, we focus on the core two-stage synthesis: person-specific dynamic parameters are first estimated idiographically, then synthesized using multivariate random-effects meta-analysis (cast as meta-analytic SEM) with full propagation of each individual's Stage 1 sampling covariance matrix. The same SEM template can be extended to incorporate covariates, distal outcomes, and other second-stage structures, but we treat these as future directions and return to them in the General Discussion.

Evaluation and Empirical Illustration

The remainder of the paper is organized to evaluate the framework's operating characteristics and demonstrate its applied utility. We first evaluate MetaVAR in a Monte Carlo simulation that varies sample size and per-person series length and focuses on recovery of population-average dynamics, between-person heterogeneity, and uncertainty quantification when Stage 1 sampling covariance matrices are propagated into Stage 2. We then illustrate the end-to-end workflow using intensive longitudinal affect data. We conclude with a general discussion that synthesizes the methodological contributions, offers practical guidance for applied researchers, and highlights limitations and directions for future work.

Monte Carlo Simulation Study

We conducted a Monte Carlo simulation study to evaluate the operating characteristics of MetaVAR under a controlled bivariate affect-dynamics data-generating process (DGP), and to compare its performance against two commonly used alternatives: a Bayesian multilevel VAR (BMLVAR) approach and a naïve "fit-many-then-summarize" approach. The simulation examined recovery of (a) population-average dynamic parameters, (b) between-person heterogeneity in those parameters, and (c) uncertainty quantification across combinations of sample size (N) and per-person time-series length (T).

Because MetaVAR is specifically designed to propagate Stage 1 sampling covariance into a multivariate second-stage synthesis, we expected its main advantages to be most pronounced for the recovery of between-person heterogeneity and the calibration of interval estimates for those heterogeneity parameters.

Data-Generating Process

Data were generated using the `SimSSMVARIVary` function from the `simStateSpace` package (v1.2.15; Pesigan et al., 2025). The data-generating model was the same centered first-order VAR model used in Stage 1 (Equation 1) for person $i = 1, \ldots, N$ and occasion $t = 1, \ldots, T$.

The person-specific set points were generated as

$$\boldsymbol{\mu}_i \sim \mathcal{N}\left(\begin{pmatrix} 2.87 \\ 2.04 \end{pmatrix}, \begin{pmatrix} 1.20 & 0.46 \\ 0.46 & 1.10 \end{pmatrix}\right),$$

and the person-specific transition matrices were generated by drawing

$$\operatorname{vec}(\mathbf{A}_i) \sim \mathcal{N}\left(\begin{pmatrix} 0.280 \\ -0.035 \\ -0.048 \\ 0.260 \end{pmatrix}, \begin{pmatrix} 0.017 & 0.005 & 0.001 & 0.008 \\ 0.005 & 0.008 & 0.001 & 0.006 \\ 0.001 & 0.001 & 0.001 & 0.002 \\ 0.008 & 0.006 & 0.002 & 0.026 \end{pmatrix}\right).$$

Here, vec$(\cdot)$ denotes the vectorization operator that stacks the columns of a matrix into a single column vector. For a 2×2 matrix $\mathbf{A}_i = \left(a_{jk,i}\right)$, this implies vec$(\mathbf{A}_i) = (a_{11,i},\, a_{21,i},\, a_{12,i},\, a_{22,i})'$.

Because the simulation allowed between-person heterogeneity only in the set points and transition parameters, the Stage 2 fixed-effect target can be written as

$$\boldsymbol{\alpha} = \left(\mathbb{E}[\boldsymbol{\mu}_i]', \mathbb{E}[\operatorname{vec}(\mathbf{A}_i)]'\right)'.$$

Likewise, the Stage 2 random-effects target is the between-person covariance matrix of $(\boldsymbol{\mu}_i', \operatorname{vec}(\mathbf{A}_i)')'$. Under the present DGP, this matrix is block diagonal because the set points and transition parameters were generated independently:

$$\boldsymbol{\tau}^2 = \begin{pmatrix} \boldsymbol{\Sigma}_\mu & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_A \end{pmatrix}.$$

This notation makes explicit that the simulation's fixed-effect targets correspond to $\boldsymbol{\alpha}$ and its heterogeneity targets correspond to $\boldsymbol{\tau}^2$, matching the labels used in the results figures below.

To ensure a stationary VAR process for each individual, generated transition matrices were retained only if all eigenvalues of $\mathbf{A}_i$ had moduli less than 1 (i.e., spectral radius < 1). In practice, this was implemented via rejection sampling using `simStateSpace`'s `SimBetaN` helper: candidate draws of $\text{vec}(\mathbf{A}_i)$ from the target multivariate normal distribution were repeatedly proposed until the corresponding $\mathbf{A}_i$ satisfied the stability constraint. Thus, the simulation targets the setting in which person-specific dynamics are heterogeneous but dynamically stable.

The process-noise covariance matrix was held constant across persons and occasions,

$$\boldsymbol{\Xi} = \begin{pmatrix} 1.30 & 0.57 \\ 0.57 & 1.56 \end{pmatrix},$$

so that the simulation isolates method performance for recovering heterogeneity in the set points and lagged dynamics. Fixing $\boldsymbol{\Xi}$ also simplifies interpretation of differences across methods because any between-person variation in estimated innovation parameters arises from finite-sample estimation error rather than true heterogeneity in the DGP.

The parameter values were informed by prior empirical work on affective dynamics (Bringmann et al., 2013), but were not copied exactly. In particular, we parameterized the person-specific *mean* (set point) $\boldsymbol{\mu}_i$ rather than the intercept vector. This choice is substantively convenient because $\boldsymbol{\mu}_i$ directly represents each person's typical equilibrium level in the centered VAR parameterization, whereas the intercept in the uncentered form is a transformed quantity (i.e., a function of both the mean and transition matrix). The average transition matrix implies moderate positive autoregressive persistence (inertia) for both variables and small negative cross-lagged effects, representing a stable bivariate regulatory system with weak reciprocal inhibitory coupling on average. The nonzero covariance among elements of $\text{vec}(\mathbf{A}_i)$ further induces realistic heterogeneity in dynamic signatures across individuals.

Design Factors

We primarily used a 3×3 factorial design crossing three sample sizes and three numbers of measurement occasions per individual:

$$N \in \{50, 100, 200\}, \qquad T \in \{50, 100, 200\}.$$

This yielded nine *balanced* conditions (Tasks 1–9), where all individuals in a given replication shared the same series length T.

In addition, to induce realistic heterogeneity in first-stage precision—and thus variability in the person-specific sampling covariance matrices used in Stage 2 (denoted $\mathbf{V}_i^2$)—we included three *unbalanced-T* conditions (Tasks 10–12). In these conditions, the sample size was fixed at $N \in \{50, 100, 200\}$ for Tasks 10, 11, and 12, respectively, but the number of measurement occasions varied across individuals within the same replication. Specifically, for each individual i, the realized series length was $T_i \in \{50, 100, 200\}$, so that within a single replication some individuals contributed shorter series (e.g., $T_i = 50$) and others contributed longer series (e.g., $T_i = 200$), producing heterogeneous $\mathbf{V}_i$ across persons. Analyses used each individual's realized T_i.

For each condition (Tasks 1–12), we generated $R = 1{,}000$ Monte Carlo replications. In each replication, person-specific time series were generated under the DGP above, analyzed by each competing method, and summarized using the performance metrics described below.

Methods Compared

We compared three approaches that represent the main inferential strategies discussed in the Introduction: (a) a two-stage method with explicit propagation of first-stage uncertainty (MetaVAR), (b) a one-step hierarchical estimator (Bayesian multilevel VAR; BMLVAR), and (c) a two-stage "fit-many-then-summarize" baseline that does not propagate first-stage uncertainty (hereafter, the naïve approach). The comparison was designed to isolate the consequences of one-step versus two-step estimation and whether uncertainty in person-specific Stage 1 estimates is carried forward into population-level inference.

Across all methods, the target of inference was the same: population-average dynamic parameters (fixed effects) and between-person heterogeneity in those parameters (random effects), as defined by the simulation DGP. To facilitate comparability, all methods were fit to the same simulated datasets under each (N, T) condition, and summaries were mapped to a common parameterization whenever possible (i.e., set points/means, VAR transition parameters, and between-person variability in those parameters). Differences across methods therefore reflect estimation strategy and inferential assumptions rather than differences in data input.

MetaVAR

MetaVAR was implemented as a two-stage procedure using `fitVARMxID` (v1.0.2) for Stage 1 and `metaDyn` (v1.0.0) for Stage 2; both packages relied on `OpenMx` (v2.22.10; Hunter, 2017; Neale et al., 2015) for model estimation. In Stage 1, a person-specific VAR(1) model was fit separately for each individual to obtain (a) a vector of estimated dynamic parameters and (b) the corresponding estimated sampling covariance matrix for that parameter vector. The Stage 1 outputs were then passed to Stage 2, where the person-level parameter vectors were synthesized using a multivariate random-effects meta-analytic model (implemented in SEM form). This Stage 2 model treats the person-specific sampling covariance matrices as known inputs, thereby propagating first-stage estimation uncertainty into the population-level likelihood.

A key inferential feature of MetaVAR is that it operates on the *full parameter vector* for each person rather than pooling parameters one at a time. Consequently, MetaVAR carries forward the within-person sampling covariance among estimated dynamic parameters into Stage 2 and estimates a coherent between-person random-effects covariance structure across the parameter system. In the simulation, we evaluated MetaVAR using model-based normal-theory confidence intervals at Stage 2 to assess finite-sample behavior.

BMLVAR

BMLVAR was implemented as a one-step hierarchical estimator using the Dynamic Structural Equation Modeling (DSEM) framework in `Mplus` (v9; Muthén & Muthén, 2017). As a one-step comparator, we fit a Bayesian multilevel VAR model to the full stacked multilevel time-series data. Under this approach, person-specific dynamics and population-level parameters are estimated jointly within a single hierarchical model. In contrast to MetaVAR, which separates idiographic estimation (Stage 1) from population synthesis (Stage 2), BMLVAR performs simultaneous estimation and naturally induces partial pooling (shrinkage) of person-specific parameters toward the population distribution.

BMLVAR was included because it represents a widely used and substantively appealing framework for multilevel intensive longitudinal data, especially when per-person time series are short and borrowing strength across individuals can stabilize estimation. At the same time, joint estimation and shrinkage make BMLVAR an informative benchmark against MetaVAR, which is intentionally designed to preserve a modular two-stage workflow while still supporting principled population inference. For BMLVAR, uncertainty was summarized using posterior credible-intervals.

Naïve

As a baseline two-stage approach, person-specific VAR(1) models were fit separately and then summarized across individuals without propagating first-stage estimation uncertainty in the second stage. Specifically, the second-stage analysis treated the Stage 1 point estimates as error-free and fit only the mean vector and covariance structure of the person-specific estimates using `lavaan` (v0.6–21; Rosseel, 2012).

This approach was included because it reflects a common practical workflow in idiographic time-series research: fit models person-by-person, extract parameters of substantive interest, and then summarize those parameters descriptively across individuals. Its advantages are modularity, transparency, and computational simplicity. However, because the second-stage summaries do not account for the within-person sampling covariance of the Stage 1 estimates (i.e., they implicitly assume $\mathbf{V}_i = \mathbf{0}$ for all individuals), the estimated between-person covariance of the Stage 1 estimates reflects both true heterogeneity and first-stage sampling variability. Consequently, random-effects variances and covariances can be inflated, particularly when per-person time series are short or Stage 1 estimation is unstable. The naïve approach therefore provides a useful baseline for quantifying the inferential gains from MetaVAR's error-propagating synthesis stage.

Comparative expectations. The three methods differ in ways that are directly relevant to the goals of the simulation. Relative to the naïve approach, MetaVAR is expected to improve population-level inference by weighting and synthesizing person-specific estimates while carrying forward their sampling uncertainty. Relative to BMLVAR, MetaVAR trades one-step joint estimation for a modular two-stage workflow that preserves idiographic estimation and supports system-level meta-analytic synthesis. The simulation therefore evaluates not only point-estimate recovery but also the inferential consequences of these distinct estimation strategies.

Performance Metrics

We evaluated performance separately for (a) population-average (fixed-effect) parameters and (b) between-person heterogeneity (random-effect) parameters.

Point-Estimate Performance

For each target parameter θ (e.g., an element of the population mean vector or heterogeneity covariance matrix), we computed Monte Carlo relative bias and RMSE across replications:

$$\text{Relative Bias}(\hat{\theta}) = \frac{\bar{\hat{\theta}} - \theta}{\theta},$$
$$\text{RMSE}(\hat{\theta}) = \sqrt{R^{-1} \sum_{r=1}^{R} \left(\hat{\theta}_r - \theta\right)^2}, \text{ where}$$
$$\bar{\hat{\theta}} = R^{-1} \sum_{r=1}^{R} \hat{\theta}_r.$$

Interval Performance (Empirical Coverage)

We evaluated uncertainty quantification using empirical coverage of nominal 95% interval estimates. For a given target parameter θ, empirical coverage was defined as

$$\widehat{\text{Coverage}}(\theta) = R^{-1} \sum_{r=1}^{R} \mathbb{I}(\theta \in \mathcal{I}_{r,.95}),$$

where $\mathcal{I}_{r,.95}$ denotes the nominal 95% interval estimate obtained in replication r.

For MetaVAR and the naïve approach, $\mathcal{I}_{r,.95}$ was the nominal 95% confidence interval. For BMLVAR, $\mathcal{I}_{r,.95}$ was the nominal 95% posterior credible interval. Although credible-interval coverage is not identical in interpretation to frequentist confidence-interval coverage, the empirical inclusion rate of nominal 95% intervals provides a practically useful operating-characteristic summary across methods.

Coverage values near .95 indicate well-calibrated interval estimation. As a secondary benchmark, Bradley's liberal robustness criterion implies that, for nominal 95% intervals, empirical coverage between approximately .92 and .97 may be regarded as acceptable (Bradley, 1978). Because the present simulation included both frequentist and Bayesian interval estimators, we refer to this metric generically as *empirical coverage of nominal 95% intervals*.

Hypothesis-Testing Performance (Empirical Power)

Because the present DGP did not include exact-zero target parameters, we evaluated inferential sensitivity using empirical power (i.e., empirical detection rates) for parameters with nonzero true values rather than empirical Type I error.

For a given target parameter θ and nominal level α (e.g., $\alpha = .05$), empirical power was defined as

$$\widehat{\text{Power}}(\theta) = R^{-1} \sum_{r=1}^{R} \mathbb{I}(\text{reject } H_0 : \theta = 0 \text{ in replication } r)\,,$$

where $\mathbb{I}(\cdot)$ is the indicator function.

For MetaVAR and the naïve approach, rejection was based on whether the nominal 95% confidence interval excluded zero. For BMLVAR, we report the empirical detection rate based on whether the nominal 95% posterior credible interval excluded zero. Although this Bayesian criterion is not a frequentist hypothesis test, it provides a practically comparable summary of inferential sensitivity across methods.

Because several true effects in the DGP were small in magnitude—especially the cross-lagged effects and some heterogeneity components—empirical power is interpreted jointly with bias, RMSE, and coverage.

Simulation Targets

The simulation primarily focused on recovery of: (a) the population-average set points and transition parameters, (b) the between-person heterogeneity in these parameters, and (c) the quality of uncertainty quantification under each method, as indexed by empirical coverage of nominal 95% interval estimates and empirical power.

This design tests MetaVAR's core claim: propagating person-specific Stage 1 sampling covariance into a multivariate second stage improves population-level inference relative to naïve aggregation while preserving the modularity of a two-stage workflow. More broadly, it addresses when such a two-stage approach approximates or departs from one-step hierarchical estimation, especially for heterogeneity recovery and interval calibration.

All code required to reproduce the simulation design, data generation, model fitting, summary metrics, and manuscript figures is available in the `manMetaVAR` repository at https://github.com/jeksterslab/manMetaVAR. The repository contains the scripts used to generate the Monte Carlo conditions, fit MetaVAR, BMLVAR, and the naïve comparator, and produce the summary results reported below.

Monte Carlo Simulation Results

Simulation performance was evaluated using relative bias, root mean square error (RMSE), empirical coverage of nominal 95% intervals, and empirical power. Coverage was interpreted relative to the nominal 95% target, with values near .95 indicating well-calibrated interval estimation; as a secondary benchmark, Bradley's liberal robustness criterion implies an acceptable range of approximately .92 to .97 for nominal 95% intervals (Bradley, 1978).

Averaged over all parameters and design cells, MetaVAR provided the most favorable overall balance of bias, RMSE, and interval calibration. Specifically, MetaVAR yielded the smallest mean absolute relative bias (|rel_bias| = 0.186), the lowest mean RMSE (0.043), and the mean coverage closest to nominal (0.930). BMLVAR was intermediate (|rel_bias| = 1.015, RMSE = 0.047, coverage = 0.760), whereas the naïve method produced the highest mean power (0.810) but also the largest mean absolute relative bias (1.540) and the poorest coverage (0.636). Thus, when results were aggregated across all targets, MetaVAR provided the strongest overall trade-off between point-estimate accuracy and interval performance.

Figures 1–8 complement these aggregate summaries by displaying parameter-by-parameter performance separately for population-average and heterogeneity parameters across all (N, T) conditions.

However, this overall pattern differed meaningfully for population-average parameters versus heterogeneity parameters. For population-average parameters, all three methods showed relatively small average bias, and BMLVAR performed best overall in interval calibration. Across fixed effects, mean absolute relative bias was small for all methods (0.005 for BMLVAR, 0.024 for MetaVAR, and 0.039 for naïve), with BMLVAR yielding the best mean coverage (0.962), followed by MetaVAR (0.910) and naïve (0.860). Mean power for fixed effects was high for all three methods (approximately 0.92). Thus, for recovery of population-average dynamics alone, BMLVAR was highly competitive and, in terms of fixed-effect coverage, performed best in the present simulation.

This fixed-effect pattern is also visible in Figures 1, 3, 5, and 7. Across most fixed-effect parameters, the three methods were fairly similar in RMSE and power, whereas the clearest differences appeared in coverage under shorter series and for the smaller cross-lagged effects.

The clearest advantage of MetaVAR emerged for the heterogeneity parameters. Across random-effects variances and covariances, MetaVAR yielded substantially smaller mean absolute relative bias (0.260) and lower mean RMSE (0.039) than either BMLVAR (1.482 and 0.047, respectively) or naïve (2.233 and 0.044, respectively). MetaVAR also showed markedly better interval calibration for heterogeneity parameters, with mean coverage of 0.939, compared with 0.666 for BMLVAR and 0.533 for naïve. This pattern is central to the motivation for MetaVAR: once the scientific target includes between-person heterogeneity in the dynamic system, carrying forward Stage 1 sampling covariance materially improves second-stage inference relative to treating person-specific estimates as error-free.

The corresponding parameter-by-parameter results are shown in Figures 2, 4, 6, and 8. These panels make clear that the main advantage of MetaVAR was concentrated in the heterogeneity parameters, for which the naïve approach showed the greatest bias and undercoverage and BMLVAR remained intermediate.

Sample size primarily affected RMSE and power. For MetaVAR, mean RMSE decreased from 0.057 at $N = 50$ to 0.030 at $N = 200$, while mean power increased from 0.615 to 0.812. BMLVAR exhibited a similar pattern, with mean RMSE decreasing from 0.066 to 0.031 and mean power increasing from 0.678 to 0.842 as sample size increased from $N = 50$ to $N = 200$. The naïve method also showed lower RMSE and higher power with larger N (RMSE: 0.061 to 0.034; power: 0.738 to 0.875), but its coverage deteriorated as sample size increased (0.741 at $N = 50$ vs. 0.529 at $N = 200$). For MetaVAR, average coverage remained relatively close to nominal across sample sizes (0.939, 0.933, and 0.916 for $N = 50$, 100, and 200, respectively), whereas BMLVAR remained consistently below nominal across all N conditions (0.783, 0.756, and 0.740).

This sample-size trend is most visible in the RMSE and power panels (Figures 3–8), where larger N generally shifted the results toward lower RMSE and higher empirical detection rates.

The number of measurement occasions had an even clearer impact on performance, especially for MetaVAR and the naïve method. For MetaVAR, mean absolute relative bias decreased from 0.475 at $T = 50$ to 0.040 at $T = 200$, while mean power increased from 0.652 to 0.779 and mean coverage improved from 0.907 to 0.939. The unbalanced-T condition performed similarly to the balanced $T = 100$ condition, with mean absolute relative bias of 0.099, RMSE of 0.042, coverage of 0.937, and power of 0.714. BMLVAR also improved as the number of occasions increased, although more modestly: mean absolute relative bias decreased from 1.207 at $T = 50$ to 0.894 at $T = 200$, RMSE decreased from 0.049 to 0.045, and power increased from 0.698 to 0.819, but coverage remained low and essentially unchanged (between 0.754 and 0.770). The naïve method showed substantial improvements in bias and coverage as T increased, with mean absolute relative bias decreasing from 2.758 at $T = 50$ to 0.601 at $T = 200$ and coverage increasing from 0.518 to 0.764, but even in the most favorable $T = 200$ condition its average coverage remained well below nominal.

The stronger role of T is evident across all four performance metrics, especially in the bias and coverage panels for both fixed and heterogeneity parameters (Figures 1, 2, 5, and 6).

The contrast between fixed effects and heterogeneity parameters became especially pronounced when considering short series. For MetaVAR, average fixed-effect coverage was below the nominal target when $T = 50$, but heterogeneity-parameter coverage remained much closer to nominal across T conditions. By contrast, the naïve method showed severe undercoverage for heterogeneity parameters throughout, especially when T was short, consistent with the expectation that ignoring Stage 1 estimation error inflates apparent between-person variability. BMLVAR, although strong for fixed effects, also showed persistent undercoverage for heterogeneity parameters, suggesting that one-step hierarchical estimation did not fully resolve recovery of the population covariance structure under the present design.

Taken together, the results suggest that MetaVAR's main advantage lies not in maximizing empirical power, but in recovering between-person heterogeneity with better-calibrated interval estimates. BMLVAR remained highly competitive for population-average dynamics, especially fixed-effect coverage, whereas the naïve approach often achieved higher apparent power at the cost of substantial bias and undercoverage. MetaVAR is therefore most attractive when the scientific target includes heterogeneity in dynamic parameters and principled uncertainty quantification is essential.

Because several heterogeneity parameters were close to zero, relative bias was naturally amplified for some components. For that reason, the RMSE and coverage results are especially important for interpreting method performance on those parameters.

Monte Carlo Simulation Discussion

The simulation supports the central rationale for MetaVAR as a two-stage method for system-level inference with propagated Stage 1 uncertainty. Its clearest advantage was recovery of between-person heterogeneity in the dynamic parameter system. Across random-effects variances and covariances, MetaVAR outperformed both BMLVAR and the naïve approach in bias and coverage, indicating that carrying the full person-specific sampling covariance matrix into Stage 2 improves inference for heterogeneity parameters.

This advantage was target-specific rather than universal. For population-average parameters, BMLVAR was highly competitive and, in the present design, yielded the best overall fixed-effect coverage. The results therefore do not suggest that MetaVAR uniformly dominates one-step hierarchical estimation; instead, they indicate that MetaVAR is especially useful when the scientific target includes the between-person covariance structure of the dynamic system.

The naïve comparator highlights the cost of ignoring first-stage uncertainty in two-stage workflows. Although it often produced relatively high empirical power, that apparent sensitivity came with substantial bias and interval undercoverage for heterogeneity parameters. This pattern is consistent with the expectation that treating person-specific estimates as error-free causes second-stage dispersion to reflect both true heterogeneity and first-stage estimation noise.

The design factors also clarify when MetaVAR is most advantageous. Increasing the number of measurement occasions improved all methods, but the gains were especially important for MetaVAR's recovery of heterogeneity parameters. This is substantively sensible: longer person-specific series improve first-stage precision, yielding more informative Stage 1 sampling covariance matrices and, in turn, more reliable Stage 2 synthesis. Short series remained challenging for all methods, particularly for small cross-lagged effects and heterogeneity components near zero.

Fourth, the design factors clarify when MetaVAR is most advantageous. Increasing the number of measurement occasions improved all methods, but the gains were especially important for MetaVAR's recovery of heterogeneity parameters. This pattern is substantively sensible: longer person-specific series improve first-stage precision, yielding more informative $\mathbf{V}_i^2$ matrices and, in turn, more reliable Stage 2 synthesis. By contrast, short series ($T = 50$) remained challenging for all methods, particularly for small cross-lagged effects and heterogeneity components near zero.

Taken together, these findings position MetaVAR as a useful two-stage option when researchers want to preserve idiographic estimation while drawing multivariate population inferences about heterogeneity in dynamic systems.

MetaVAR should not be interpreted as a universal replacement for one-step hierarchical modeling. Rather, its contribution is to offer a principled two-stage middle ground: one that preserves idiographic estimation, scales more naturally to modular and potentially complex first-stage workflows, and still supports multivariate population synthesis with full error propagation. The present results suggest that this middle ground is especially useful when researchers care about how dynamic parameters vary across people, not merely what their average values are.

At the same time, several limitations of the simulation should be noted. The DGP was bivariate, the innovation covariance was fixed across persons, and all data were generated from the same model class used for estimation. Future work should extend the evaluation to higher-dimensional systems, heterogeneity in innovation processes, model misspecification, missing data, irregular spacing, and Stage 1 models beyond observed-variable VARs. These extensions are important because they speak to the broader claim that MetaVAR can serve as a general two-stage synthesis framework for person-specific dynamical models.

Empirical Illustration: Meta-Analysis of Daily Affect Dynamics

We illustrate the proposed MetaVAR workflow using intensive longitudinal data from the Affective Dynamics and Individual Differences study (ADID; Emotions & Dynamic Systems Laboratory, 2010), which was designed to examine how emotion dynamics unfold in everyday life and how these dynamics differ across individuals. The ADID EMA protocol sampled participants five times per day for approximately 30 days (four daytime prompts plus an evening report), yielding up to 150 measurement occasions per participant in the raw data. ADID has been published in part across multiple substantive and methodological papers, including work on regime-switching dynamics, measurement/trait–state structure, network and VAR-based modeling, outlier-robust state-space modeling, and missing-data methods (Chow & Zhang, 2013; Gates et al., 2023; Guo et al., 2012; Hutton & Chow, 2014; Li et al., 2022, 2024; Oh et al., 2025; Ou et al., 2023; Park et al., 2020; You et al., 2019).

This design directly matches the core goal of MetaVAR: estimating person-specific dynamic "signatures" in a first stage and then synthesizing them at the population level in a second stage while propagating each individual's Stage 1 sampling uncertainty. In the present illustration, we focus on within-person dynamics between negative affect (NA) and positive affect (PA). Specifically, we meta-analyze the full person-specific parameter system consisting of (a) affective set points, (b) temporal dynamics encoded by the VAR(1) transition matrix—autoregressive (AR) inertia and cross-regressive (CR) coupling—and (c) the process-noise (innovation) covariance matrix, which captures within-occasion co-fluctuation of unexpected changes after accounting for lagged dynamics.

Substantively, we focus on NA and PA because emotion-dynamics theories treat affect as a self-regulating system characterized by (at least) a typical level (often conceptualized as a "set point") and temporal persistence from one occasion to the next (often conceptualized as affective "inertia"), with potential cross-lagged coupling between NA and PA reflecting carryover and regulatory spillover (Houben et al., 2015; Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). These features are of substantive interest because they capture how quickly individuals return to (or remain near) their typical affective state after perturbations. MetaVAR treats heterogeneity in these features—including heterogeneity in the innovation covariance structure—as a scientific target, providing population-average inferences while quantifying reliable between-person variation and co-variation across elements of the dynamic system.

Measures

Momentary affect in ADID was assessed using adjective ratings drawn from established affect instruments, including items from the Positive Affect and Negative Affect Schedule (PANAS; Watson et al., 1988) and circumplex-content items used to broaden coverage of activation levels (Russell, 1980). Participants rated how often they experienced each affect descriptor since the previous assessment on an ordinal Likert-type scale, as implemented in the ADID daily diary surveys. Using the recorded date–time stamps, we rounded responses to the nearest hour and then constructed an hourly, regularly spaced time series by inserting missing values for hours with no report. Following prior ADID applications, we construct per-occasion NA and PA composites by averaging the available negative-affect and positive-affect items, respectively. To reduce the influence of slow drift over the month-long protocol and to better align the series with the stationary VAR(1) assumption, we apply a within-person linear detrending step to each affect composite prior to model fitting. In the VAR(1) model, person-specific set points represent typical affect levels, the transition matrix encodes temporal persistence and cross-lagged coupling, and the innovation covariance matrix characterizes same-occasion co-fluctuation of unexpected deviations after accounting for lagged dynamics.

Results

Stage 1 estimated person-specific NA–PA dynamics for each participant, and Stage 2 synthesized these individual "dynamic signatures" to obtain (a) population-average dynamics, (b) between-person heterogeneity in these dynamics, and (c) systematic co-variation among elements of the parameter system. Below we summarize the main results in this order.

Stage 1: Person-Specific VAR

We fit the person-specific VAR(1) dynamics model in Equation 1 separately for each participant, using the `fitVARMxID` package (v1.0.2), estimating all dynamic parameters as individual-specific (i.e., person-specific set points, autoregressive coefficients, cross-regressive coefficients, and innovation covariance matrix). Models were fit for $N = 217$ individuals.

Stage 2: Multivariate Random-Effects Meta-Analysis of the Dynamic Parameter System

Stage 2 synthesized the person-specific VAR(1) parameter systems using a multivariate random-effects meta-analysis with the `metaDyn` package (v1.0.0). The meta-analytic mean vector $\boldsymbol{\alpha}$ contained nine elements, ordered as: the two set points $(\mu_{\text{NA}}, \mu_{\text{PA}})$, the four VAR transition coefficients $(\beta_{11}, \beta_{21}, \beta_{12}, \beta_{22})$ corresponding to AR and cross-regressive (CR) effects, and the three unique elements of the innovation covariance matrix $\boldsymbol{\Psi}$ for NA and PA $(\psi_{11}, \psi_{21}, \psi_{22})$.

Population-average dynamics ($\boldsymbol{\alpha}$). The population-average set points indicated higher typical PA than NA: $\mu_{\text{NA}} = 1.4401$ (SE = 0.0253), 95% CI [1.3906, 1.4897], and $\mu_{\text{PA}} = 2.5817$ (SE = 0.0322), 95% CI [2.5187, 2.6448]. The average temporal dynamics showed positive inertia in both affect dimensions, with $\beta_{11} = 0.3610$ (SE = 0.0309), 95% CI [0.3004, 0.4216] for NA and $\beta_{22} = 0.3194$ (SE = 0.0313), 95% CI [0.2581, 0.3807] for PA. Cross-regressive effects were negative on average. The NA→PA effect was $\beta_{21} = -0.0678$ (SE = 0.0358), 95% CI [-0.1380, 0.0024], which was marginal and not conventionally distinguishable from zero ($p = .0585$). The PA→NA effect was $\beta_{12} = -0.0541$ (SE = 0.0185), 95% CI [-0.0904, -0.0178], indicating that higher PA at one occasion predicted lower subsequent NA on average ($p = .0035$).

Stage 2 also estimated the population-average innovation covariance matrix in the original metric with $\psi_{11} = 0.0333$ (SE = 0.0019), 95% CI [0.0296, 0.0370], $\psi_{21} = -0.0075$ (SE = 0.0010), 95% CI [-0.0094, -0.0056], and $\psi_{22} = 0.0612$ (SE = 0.0029), 95% CI [0.0554, 0.0670]. Thus, after accounting for lagged dynamics, the innovation variance was larger for PA than for NA, and the innovation covariance was negative, indicating that within-occasion shocks to NA and PA tended to move in opposite directions on average.

Between-person heterogeneity and co-variation ($\boldsymbol{\tau}^2$). The estimated random-effects covariance matrix $\boldsymbol{\tau}^2$ indicated substantial heterogeneity across the full parameter system. The diagonal elements (between-person variances) were 0.1373 for μ_{NA} (SD $\approx$ 0.37), 0.2227 for μ_{PA} (SD $\approx$ 0.47), 0.1837 for β_{11} (SD $\approx$ 0.43), 0.2175 for β_{21} (SD $\approx$ 0.47), 0.0591 for β_{12} (SD $\approx$ 0.24), 0.1891 for β_{22} (SD $\approx$ 0.43), and, for the innovation-covariance parameters, 0.0007 for ψ_{11} (SD $\approx$ 0.027), 0.0001 for ψ_{21} (SD $\approx$ 0.010), and 0.0016 for ψ_{22} (SD $\approx$ 0.040). Several, though not all, off-diagonal elements were also statistically distinguishable from zero, indicating systematic co-variation among components of the dynamic signature across individuals. For example, individuals with higher NA set points tended to show greater NA inertia (cov = 0.0268, $p = .0222$); NA inertia β_{11} covaried negatively with the NA→PA coupling β_{21} (cov = -0.0677, $p < .001$); and PA inertia β_{22} covaried negatively with the PA→NA coupling β_{12} (cov = -0.0225, $p = .0117$). There was also evidence of co-variation between set points and innovation parameters (e.g., μ_{NA} with ψ_{11} and ψ_{22}; both $p < .05$, and μ_{PA} with ψ_{11}, ψ_{21}, and ψ_{22}; all $p < .05$). In addition, several covariances among the innovation-covariance parameters were statistically significant (e.g., ψ_{11} with ψ_{21}, ψ_{11} with ψ_{22}, and ψ_{21} with ψ_{22}; all $p < .001$), consistent with meaningful between-person differences in the magnitude and co-fluctuation of within-occasion shocks.

Heterogeneity index ($\mathbf{I}^2$). The $\mathbf{I}^2$ indices were uniformly high for all nine elements of $\boldsymbol{\alpha}$, indicating that the observed dispersion in Stage 1 parameter estimates was overwhelmingly attributable to true between-person heterogeneity rather than within-person sampling variability. Specifically, $\mathbf{I}^2$ was 1.0000 for μ_{NA}, 0.9997 for μ_{PA}, 0.9909 for β_{11}, 0.9782 for β_{21}, 0.9934 for β_{12}, 0.9879 for β_{22}, 0.9999 for ψ_{11}, 0.9997 for ψ_{21}, and 0.9999 for ψ_{22}. Taken together, the $\boldsymbol{\tau}^2$ and $\mathbf{I}^2$ results support the core premise motivating MetaVAR: affective set points, temporal dynamics (AR/CR), and innovation covariance structure all vary reliably across individuals, making multivariate synthesis of the full parameter system scientifically informative.

Empirical Example Discussion

This empirical illustration demonstrates how MetaVAR can be used to (a) estimate idiographic affect-dynamics parameters from intensive longitudinal data and (b) synthesize these person-specific “dynamic signatures” into population-average inferences while quantifying heterogeneity (Emotions & Dynamic Systems Laboratory, 2010; Fisher et al., 2017). In ADID, the approach yielded coherent population-average dynamics, strong evidence of between-person heterogeneity in both set points and lagged dependencies, and interpretable co-variation among elements of the parameter system (Houben et al., 2015). In addition to the familiar *temporal* dependencies (AR and CR effects in the VAR transition matrix), MetaVAR also supports population inference on the *contemporaneous* layer implied by the process innovations—summarized here via the innovation covariance matrix—which captures same-occasion co-fluctuation of unexpected changes after accounting for lagged dynamics.

Population-Average NA–PA Dynamics

At the population level, participants reported higher typical PA than NA, consistent with findings from intensive longitudinal studies of community samples in which average affective tone tends to be more positive than negative (Houben et al., 2015; Watson et al., 1988). Both NA and PA exhibited positive inertia, with AR(NA) slightly larger than AR(PA), suggesting that momentary elevations in NA tended to persist somewhat more strongly than momentary elevations in PA at the hourly scale used here (Houben et al., 2015; Kuppens, Oravecz, & Tuerlinckx, 2010). Cross-lagged coupling was negative in both directions. Higher PA predicted lower subsequent NA on average, whereas the NA→PA effect was also negative but only marginally distinguishable from zero. Interpreted as a bivariate self-regulating system, these average cross-lagged effects are broadly consistent with reciprocal inhibition or spillover processes discussed in affect-dynamics theory (Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). Importantly, these are descriptive features at a specific time scale (hourly, after within-person detrending) and should be interpreted as average short-term dependencies rather than mechanistic causal effects (Asparouhov & Muthén, 2018; Hamaker & Grasman, 2015).

The population-average innovation covariance matrix further indicated that, after accounting for lagged dynamics, within-occasion unexpected shifts in NA and PA tended to move in opposite directions on average. At the same time, the larger innovation variance for PA than for NA suggests that short-term unexplained fluctuations were greater for PA in this dataset.

Heterogeneity as a Substantive Finding, Not a Nuisance

The random-effects variance components indicated substantial between-person variability across the dynamic signature, including both set points and lagged dynamics as well as the innovation covariance parameters. The corresponding $\mathbf{I}^2$ values were uniformly high, implying that the observed dispersion in Stage 1 estimates overwhelmingly reflects true between-person differences rather than sampling variability of the person-specific estimates (Higgins & Thompson, 2002). Methodologically, these results motivate multivariate synthesis rather than edge-by-edge pooling, because heterogeneity was present across parameters and the off-diagonal elements of $\boldsymbol{\tau}^2$ revealed how deviations in set points, inertia, coupling, and innovation structure co-occurred across individuals (Cheung, 2015).

Several examples illustrate this point. Individuals with higher NA set points tended to show greater NA inertia, suggesting that typical affective level and short-term persistence may be linked within persons. Likewise, stronger NA inertia was associated with more negative NA→PA coupling, and stronger PA inertia was associated with more negative PA→NA coupling, indicating that temporal persistence and cross-affect coupling were not independent sources of heterogeneity. Co-variation among the innovation-covariance parameters further suggests meaningful between-person differences in the magnitude and structure of within-occasion affective shocks.

Practical and Methodological Considerations

Several features of the analysis highlight common trade-offs in applied dynamic modeling. First, a small fraction of Stage 1 models did not converge, underscoring the importance of diagnostics and robustness checks in idiographic estimation (e.g., sensitivity to starting values, stationarity constraints, alternative innovation structures, or mild regularization) (Gates et al., 2023; Ou et al., 2023). Second, we imposed regular spacing by rounding to the nearest hour and inserting missing values, and we applied within-person linear detrending to better align the data with a stationary VAR(1) approximation; both steps improve feasibility but may also alter the interpretation of parameters (e.g., detrending shifts focus from slower drifts to shorter-term fluctuations) (Guo et al., 2012; Li et al., 2022). Third, NA/PA composites were constructed from ordinal item ratings and treated as approximately continuous; future work could examine whether a latent measurement model or item-level modeling changes the estimated dynamic signature and its heterogeneity (Park et al., 2020; Watson et al., 1988).

Implications and Future Directions

Overall, the illustration supports the substantive value of treating affect dynamics as person-specific signatures that can be meaningfully compared and synthesized (Fisher et al., 2017; Houben et al., 2015). Several extensions follow naturally from the same Stage 2 SEM template but are not pursued in the present illustration, including (a) covariate moderation of the parameter system and (b) linking dynamic signatures to distal outcomes, both under full Stage 1 error propagation. Additional directions include allowing nonstationarity (e.g., regime switching or time-varying parameters; Chow & Zhang, 2013; Hutton & Chow, 2014), extending Stage 1 to continuous-time formulations to better handle irregular sampling without discretization (Park et al., 2025), and targeting multivariate constraints or derived functionals (e.g., stability summaries, impulse-response metrics, or innovation-based summaries) as higher-level quantities for synthesis.

General Discussion and Future Directions

The central aim of this paper was to develop a principled way to move from person-specific dynamic models to population-level inference without collapsing the distinction between within-person and between-person structure. MetaVAR addresses this aim by combining idiographic estimation in Stage 1 with multivariate meta-analytic structural equation modeling in Stage 2. This yields a middle-ground workflow between one-step hierarchical estimation and uncorrected fit-many-then-summarize approaches: it preserves person-specific modeling, propagates first-stage uncertainty, and supports system-level inference on the full joint parameter vector.

More broadly, these findings clarify where MetaVAR fits in the methodological landscape. It is not a universal replacement for one-step hierarchical modeling; rather, it offers a principled two-stage middle ground that preserves idiographic estimation while supporting multivariate population synthesis with full error propagation. In this sense, MetaVAR extends the idiographic two-stage perspective of Lee and Gates (2023) to multivariate dynamic systems, where the scientific target is often the joint parameter system rather than isolated paths. This perspective also aligns with the broader meta-analytic view that average effects should be interpreted alongside heterogeneity, not in place of it.

Several contributions follow from this framework. First, MetaVAR formalizes multilevel dynamic modeling as a meta-analytic problem in which persons play the role of studies. This makes it possible to synthesize not only average dynamics, but also between-person heterogeneity and cross-parameter co-variation across the dynamic system. Second, because Stage 2 is expressed as an SEM, MetaVAR naturally supports model-level constraints, multivariate hypotheses, and extensions involving moderators or distal outcomes. Third, the framework is not limited in principle to observed-variable VAR models. Although the present manuscript focuses on VAR(1) dynamics, the same two-stage logic extends to any Stage 1 model that yields a parameter vector and an accompanying sampling covariance matrix.

The Monte Carlo results clarify where this framework is most useful. MetaVAR did not uniformly dominate the comparison methods across every target. For population-average parameters, the Bayesian multilevel VAR comparator was highly competitive and, in the present design, performed especially well in fixed-effect coverage. The clearest advantage of MetaVAR instead emerged for the recovery of between-person heterogeneity and the calibration of interval estimates for heterogeneity parameters. This is substantively important because many scientific questions in intensive longitudinal research concern not only what dynamic processes look like on average, but also how and why those processes differ across people.

The empirical illustration further underscored this point. In the ADID data, MetaVAR recovered interpretable population-average affect dynamics while also revealing substantial heterogeneity in set points, autoregressive persistence, cross-lagged coupling, and innovation structure. These results illustrate the scientific value of treating a person's dynamic model as a multivariate signature that can be compared, related, and synthesized at the population level. More broadly, the illustration shows that heterogeneity is not merely a nuisance to be averaged away, but can itself be a target of explanation and theory building.

Our small-scale benchmarking results also highlight an important practical advantage of the MetaVAR workflow: computational scalability. Using a benchmark dataset with $N = 200$ individuals and $T = 200$ measurement occasions per individual, the Naive approach was the fastest method, MetaVAR occupied an intermediate position, and the joint Bayesian multilevel VAR approach was the slowest. Mean runtimes were approximately 11.08 seconds for Naive, 77.75 seconds for MetaVAR, and 363.79 seconds for BMLVAR. Thus, although MetaVAR incurs additional computation relative to a simple two-stage summary approach, it remained substantially faster than joint Bayesian estimation in this setting. This advantage is especially relevant for applications with large numbers of individuals, because Stage 1 idiographic estimation is embarrassingly parallel: person-specific models can be fit independently across cores or computing nodes. As a result, the framework is well suited not only to larger N settings, but also to settings in which person-specific estimation becomes more costly, such as when each individual contributes longer time series or more complex dynamic models. Additional details are provided in the benchmarking vignette at https://jeksterslab.github.io/manMetaVAR/articles/benchmark.html

At the same time, several limitations should be acknowledged. The simulation considered a bivariate data-generating process, fixed innovation covariance across persons, and correct model specification. The empirical illustration relied on observed composites, regularized hourly spacing, and a stationary VAR(1) approximation after detrending. Future work should examine higher-dimensional systems, heterogeneity in innovation processes, missing data and irregular spacing, misspecification, and Stage 1 models beyond observed-variable VARs, including latent-variable, state-space, continuous-time, and nonlinear formulations. It will also be important to study the behavior of MetaVAR when Stage 1 sampling covariances are themselves approximate, such as when obtained from sandwich estimators or bootstrap procedures.

Several extensions of the present framework are especially promising. One is moderation: person-level covariates can be incorporated in Stage 2 to explain why some individuals exhibit stronger inertia, weaker regulation, or different innovation structures than others. Another is distal-outcome modeling, in which dynamic signatures are related to clinically or developmentally meaningful endpoints. A third is the synthesis of derived functionals rather than raw parameters, such as stability summaries, impulse-response quantities, network summaries, or other theory-relevant features of the dynamic system. These directions would further strengthen the role of MetaVAR as a general synthesis framework for idiographic dynamical models.

A practical strength of the framework is that it is already implementable in accessible software. Stage 1 idiographic estimation can be carried out with the `fitVARMxID` package, which provides a convenient interface for fitting person-specific discrete-time and continuous-time VAR models in `OpenMx`; it is available on CRAN and includes support for multi-core computation to help accelerate fitting across individuals. Stage 2 synthesis can be conducted with the `metaDyn` package, which implements MetaVAR and related extensions in an `OpenMx`-based workflow. It is also available on CRAN. Together, these tools make the proposed framework more readily usable in applied intensive longitudinal research.

Overall, MetaVAR contributes a principled and flexible framework for drawing population-level conclusions from person-specific dynamics. It preserves the conceptual advantages of idiographic modeling while addressing a key inferential problem that arises once researchers wish to generalize beyond a set of individual models. For intensive longitudinal research, this provides a practical route to asking not only what is typical, but also how people differ, how those differences cohere across the parameter system, and how dynamic signatures relate to broader substantive outcomes.

References

Arbuckle, J. L. (1996). Full information estimation in the presence of incomplete data. In G. A. Marcoulides & R. E. Schumacker (Eds.), *Advanced structural equation modeling*. https://doi.org/10.4324/9781315827414

Asparouhov, T., Hamaker, E. L., & Muthén, B. (2018). Dynamic structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, *25*(3), 359–388. https://doi.org/10.1080/10705511.2017.1406803

Asparouhov, T., & Muthén, B. (2018). Latent variable centering of predictors and mediators in multilevel and time-series models. *Structural Equation Modeling: A Multidisciplinary Journal*, *26*(1), 119–142. https://doi.org/10.1080/10705511.2018.1511375

Baek, E., Luo, W., & Lam, K. H. (2023). Meta-analysis of single-case experimental design using multilevel modeling. *Behavior Modification*, *47*(6), 1546–1573. https://doi.org/10.1177/01454455221144034

Beltz, A. M., Wright, A. G. C., Sprague, B. N., & Molenaar, P. C. M. (2016). Bridging the nomothetic and idiographic approaches to the analysis of clinical data. *Assessment*, *23*(4), 447–458. https://doi.org/10.1177/1073191116648209

Berkey, C. S., Hoaglin, D. C., Antczak-Bouckoms, A., Mosteller, F., & Colditz, G. A. (1998). Meta-analysis of multiple outcomes by regression with random effects. *Statistics in Medicine*, *17*(22), 2537–2550. https://doi.org/10.1002/(sici)1097-0258(19981130)17:22<2537::aid-sim953>3.0.co;2-c

Bolger, N., Davis, A., & Rafaeli, E. (2003). Diary methods: Capturing life as it is lived. *Annual Review of Psychology*, *54*(1), 579–616. https://doi.org/10.1146/annurev.psych.54.101601.145030

Bolger, N., & Laurenceau, J.-P. (2013). *Intensive longitudinal methods: An introduction to diary and experience sampling research*. The Guilford Press.

Bollen, K. A. (1989). *Structural equations with latent variables*. Wiley. https://doi.org/10.1002/9781118619179

Bradley, J. V. (1978). Robustness? *British Journal of Mathematical and Statistical Psychology*, *31*(2), 144–152. https://doi.org/10.1111/j.2044-8317.1978.tb00581.x

Bringmann, L. F., Pe, M. L., Vissers, N., Ceulemans, E., Borsboom, D., Vanpaemel, W., Tuerlinckx, F., & Kuppens, P. (2016). Assessing temporal emotion dynamics using networks. *Assessment*, *23*(4), 425–435. https://doi.org/10.1177/1073191116645909

Bringmann, L. F., Vissers, N., Wichers, M., Geschwind, N., Kuppens, P., Peeters, F., Borsboom, D., & Tuerlinckx, F. (2013). A network approach to psychopathology: New insights into clinical longitudinal data. *PLoS ONE*, *8*(4). https://doi.org/10.1371/journal.pone.0060188

Cheung, M. W.-L. (2008). A model for integrating fixed-, random-, and mixed-effects meta-analyses into structural equation modeling. *Psychological Methods*, *13*(3), 182–202. https://doi.org/10.1037/a0013163

Cheung, M. W.-L. (2013). Multivariate meta-analysis as structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, *20*(3), 429–454. https://doi.org/10.1080/10705511.2013.797827

Cheung, M. W.-L. (2015). *Meta-analysis: A structural equation modeling approach*. Wiley. https://doi.org/10.1002/9781118957813

Chow, S.-M., & Zhang, G. (2013). Nonlinear regime-switching state-space (RSSS) models. *Psychometrika*, *78*(4), 740–768. https://doi.org/10.1007/s11336-013-9330-8

Csikszentmihalyi, M., & Larson, R. (1987). Validity and reliability of the experience-sampling method. *The Journal of Nervous and Mental Disease*, *175*(9), 526–536. https://doi.org/10.1097/00005053-198709000-00004

Curran, P. J., & Bauer, D. J. (2011). The disaggregation of within-person and between-person effects in longitudinal models of change. *Annual Review of Psychology*, *62*(1), 583–619. https://doi.org/10.1146/annurev.psych.093008.100356

Declercq, L., Jamshidi, L., Fernández Castilla, B., Moeyaert, M., Beretvas, S. N., Ferron, J. M., & Van den Noortgate, W. (2020). Multilevel meta-analysis of individual participant data of single-case experimental designs: One-stage versus two-stage methods. *Multivariate Behavioral Research*, *57*(2-3), 298–317. https://doi.org/10.1080/00273171.2020.1822148

Efron, B., & Morris, C. (1975). Data analysis using Stein's estimator and its generalizations. *Journal of the American Statistical Association*, *70*(350), 311–319. https://doi.org/10.1080/01621459.1975.10479864

Emotions & Dynamic Systems Laboratory. (2010). *The affective dynamics and individual differences (ADID) study: Developing non-stationary and network-based methods for modeling the perception and physiology of emotions*. University of North Carolina at Chapel Hill. Chapel Hill, NC.

Enders, C. K. (2010). *Applied missing data analysis*. Guilford Publications.

Epskamp, S., Waldorp, L. J., M ottus, R., & Borsboom, D. (2018). The Gaussian graphical model in cross-sectional and time-series data. *Multivariate Behavioral Research*, *53*(4), 453–480. https://doi.org/10.1080/00273171.2018.1454823

Fisher, A. J., Medaglia, J. D., & Jeronimus, B. F. (2018). Lack of group-to-individual generalizability is a threat to human subjects research. *Proceedings of the National Academy of Sciences*, *115*(27). https://doi.org/10.1073/pnas.1711978115

Fisher, A. J., Newman, M. G., & Molenaar, P. C. M. (2011). A quantitative method for the analysis of nomothetic relationships between idiographic structures: Dynamic patterns create attractor states for sustained posttreatment change. *Journal of Consulting and Clinical Psychology*, *79*(4), 552–563. https://doi.org/10.1037/a0024069

Fisher, A. J., Reeves, J. W., Lawyer, G., Medaglia, J. D., & Rubel, J. A. (2017). Exploring the idiographic dynamics of mood and anxiety via network analysis. *Journal of Abnormal Psychology*, *126*(8), 1044–1056. https://doi.org/10.1037/abn0000311

Gates, K. M., Chow, S.-M., & Molenaar, P. C. M. (2023). *Intensive longitudinal analysis of human processes*. Chapman & Hall/CRC Press, https://doi.org/10.1201/9780429172649

Gelman, A., & Hill, J. (2006). *Data analysis using regression and multilevel/hierarchical models*. Cambridge University Press. https://doi.org/10.1017/cbo9780511790942

Guo, R., Zhu, H., Chow, S., & Ibrahim, J. G. (2012). Bayesian lasso for semiparametric structural equation models. *Biometrics*, *68*(2), 567–577. https://doi.org/10.1111/j.1541-0420.2012.01751.x

Hamaker, E. L., Asparouhov, T., Brose, A., Schmiedek, F., & Muthén, B. (2018). At the frontiers of modeling intensive longitudinal data: Dynamic structural equation models for the affective measurements from the COGITO study. *Multivariate Behavioral Research*, *53*(6), 820–841. https://doi.org/10.1080/00273171.2018.1446819

Hamaker, E. L., Dolan, C. V., & Molenaar, P. C. M. (2005). Statistical modeling of the individual: Rationale and application of multivariate stationary time series analysis. *Multivariate Behavioral Research*, *40*(2), 207–233. https://doi.org/10.1207/s15327906mbr4002_3

Hamaker, E. L., & Grasman, R. P. P. P. (2015). To center or not to center? Investigating inertia with a multilevel autoregressive model. *Frontiers in Psychology*, *5*. https://doi.org/10.3389/fpsyg.2014.01492

Hamilton, J. D. (1994). *Time series analysis*. Princeton University Press.

Hedges, L. V., & Olkin, I. (1985). *Statistical methods for meta-analysis*. Academic Press.

Higgins, J. P. T., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, *21*(11), 1539–1558. https://doi.org/10.1002/sim.1186

Houben, M., Van Den Noortgate, W., & Kuppens, P. (2015). The relation between short-term emotion dynamics and psychological well-being: A meta-analysis. *Psychological Bulletin*, *141*(4), 901–930. https://doi.org/10.1037/a0038822

Hunter, M. D. (2017). State space modeling in an open source, modular, structural equation modeling environment. *Structural Equation Modeling: A Multidisciplinary Journal*, *25*(2), 307–324. https://doi.org/10.1080/10705511.2017.1369354

Hutton, R. S., & Chow, S.-M. (2014). Longitudinal multi-trait-state-method model using ordinal data. *Multivariate Behavioral Research*, *49*(3), 269–282. https://doi.org/10.1080/00273171.2014.903832

Jackson, D., White, I. R., & Riley, R. D. (2012). Quantifying the impact of between-study heterogeneity in multivariate meta-analyses. *Statistics in Medicine*, *31*(29), 3805–3820. https://doi.org/10.1002/sim.5453

Kuppens, P., Allen, N. B., & Sheeber, L. B. (2010). Emotional inertia and psychological maladjustment. *Psychological Science*, *21*(7), 984–991. https://doi.org/10.1177/0956797610372634

Kuppens, P., Oravecz, Z., & Tuerlinckx, F. (2010). Feelings change: Accounting for individual differences in the temporal dynamics of affect. *Journal of Personality and Social Psychology*, *99*(6), 1042–1060. https://doi.org/10.1037/a0020962

Kurtzer, G. M., cclerget, Bauer, M. W., Kaneshiro, I., Trudgian, D., & Godlove, D. (2021). hpcng/singularity: Singularity 3.7.3. https://doi.org/10.5281/ZENODO.1310023

Kurtzer, G. M., Sochat, V., & Bauer, M. W. (2017). Singularity: Scientific containers for mobility of compute. *PLOS ONE*, *12*(5), e0177459. https://doi.org/10.1371/journal.pone.0177459

Lee, S. A. W., & Gates, K. M. (2023). From the individual to the group: Using idiographic analyses and two-stage random effects meta-analysis to obtain population level inferences for within-person processes. *Multivariate Behavioral Research*, *59*(6), 1220–1239. https://doi.org/10.1080/00273171.2023.2229310

Li, Y., Oravecz, Z., Ji, L., & Chow, S.-M. (2024). Multiple imputation with factor scores: A practical approach for handling simultaneous missingness across items in longitudinal designs. *Multivariate Behavioral Research*, *60*(1), 61–89. https://doi.org/10.1080/00273171.2024.2371816

Li, Y., Wood, J., Ji, L., Chow, S.-M., & Oravecz, Z. (2022). Fitting multilevel vector autoregressive models in Stan, JAGS, and Mplus. *Structural Equation Modeling: A Multidisciplinary Journal*, *29*(3), 452–475. https://doi.org/10.1080/10705511.2021.1911657

Lütkepohl, H. (2005). *New introduction to multiple time series analysis*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-27752-1

Mehta, P. D., & Neale, M. C. (2005). People are variables too: Multilevel structural equations modeling. *Psychological Methods*, *10*(3), 259–284. https://doi.org/10.1037/1082-989x.10.3.259

Moeyaert, M., Ugille, M., Natasha Beretvas, S., Ferron, J., Bunuan, R., & Van den Noortgate, W. (2016). Methods for dealing with multiple outcomes in meta-analysis:a comparison between averaging effect sizes, robust variance estimation and multilevel meta-analysis. *International Journal of Social Research Methodology*, *20*(6), 559–572. https://doi.org/10.1080/13645579.2016.1252189

Molenaar, P. C. M. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement: Interdisciplinary Research & Perspective*, *2*(4), 201–218. https://doi.org/10.1207/s15366359mea0204_1

Muthén, L. K., & Muthén, B. O. (2017). *Mplus user's guide. Eighth edition*. Los Angeles, CA, Muthén & Muthén.

Neale, M. C., Hunter, M. D., Pritikin, J. N., Zahery, M., Brick, T. R., Kirkpatrick, R. M., Estabrook, R., Bates, T. C., Maes, H. H., & Boker, S. M. (2015). OpenMx 2.0: Extended structural equation and statistical modeling. *Psychometrika*, *81*(2), 535–549. https://doi.org/10.1007/s11336-014-9435-8

Nesselroade, J. R. (1991). The warp and the woof of the developmental fabric. In R. M. Downs, L. S. Liben, & D. S. Palermo (Eds.), *Visions of aesthetics, the environment & development: The legacy of Joachim F. Wohlwill* (pp. 213–240). Lawrence Erlbaum Associates, Inc.

Oh, H., Hunter, M. D., & Chow, S.-M. (2025). Measurement model misspecification in dynamic structural equation models: Power, reliability, and other considerations. *Structural Equation Modeling: A Multidisciplinary Journal*, *32*(3), 511–528. https://doi.org/10.1080/10705511.2025.2452884

Ou, L., Hunter, M. D., Lu, Z., Stifter, C. A., & Chow, S. (2023). Estimation of nonlinear mixed-effects continuous-time models using the continuous-discrete extended Kalman filter. *British Journal of Mathematical and Statistical Psychology*, *76*(3), 462–490. https://doi.org/10.1111/bmsp.12318

Park, J. J., Chow, S.-M., Fisher, Z. F., & Molenaar, P. C. M. (2020). Affect and personality: Ramifications of modeling (non-)directionality in dynamic network models. *European Journal of Psychological Assessment*, *36*(6), 1009–1023. https://doi.org/10.1027/1015-5759/a000612

Park, J. J., Fisher, Z. F., Hunter, M. D., Shenk, C., Russell, M., Molenaar, P. C. M., & Chow, S.-M. (2025). Unsupervised model construction in continuous-time. *Structural Equation Modeling: A Multidisciplinary Journal*, *32*(3), 377–399. https://doi.org/10.1080/10705511.2024.2429544

Pastor, D. A., & Lazowski, R. A. (2017). On the multilevel nature of meta-analysis: A tutorial, comparison of software programs, and discussion of analytic choices. *Multivariate Behavioral Research*, *53*(1), 74–89. https://doi.org/10.1080/00273171.2017.1365684

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. https://doi.org/10.1037/met0000779

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2). https://doi.org/10.18637/jss.v048.i02

Rowland, Z., & Wenzel, M. (2020). Mindfulness and affect-network density: Does mindfulness facilitate disengagement from affective experiences in daily life? *Mindfulness*, *11*(5), 1253–1266. https://doi.org/10.1007/s12671-020-01335-4

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, *39*(6), 1161–1178. https://doi.org/10.1037/h0077714

Shadish, W. R., Rindskopf, D. M., & Hedges, L. V. (2008). The state of the science in the meta-analysis of single-case experimental designs. *Evidence-Based Communication Assessment and Intervention*, *2*(3), 188–196. https://doi.org/10.1080/17489530802581603

Shiffman, S., Stone, A. A., & Hufford, M. R. (2008). Ecological momentary assessment. *Annual Review of Clinical Psychology*, *4*(1), 1–32. https://doi.org/10.1146/annurev.clinpsy.3.022806.091415

Sliwinski, M. J. (2008). Measurement-burst designs for social health research. *Social and Personality Psychology Compass*, *2*(1), 245–261. https://doi.org/10.1111/j.1751-9004.2007.00043.x

Tange, O. (2021). GNU Parallel 20210922 ('Vindelev') [stable]. https://doi.org/10.5281/ZENODO.5523272

Trull, T. J., & Ebner-Priemer, U. (2013). Ambulatory assessment. *Annual Review of Clinical Psychology*, *9*(1), 151–176. https://doi.org/10.1146/annurev-clinpsy-050212-185510

Van den Noortgate, W., & Onghena, P. (2008). A multilevel meta-analysis of single-subject experimental design studies. *Evidence-Based Communication Assessment and Intervention*, *2*(3), 142–151. https://doi.org/10.1080/17489530802505362 van Houwelingen, H. C., Arends, L. R., & Stijnen, T. (2002). Advanced methods in meta-analysis: Multivariate approach and meta-regression. *Statistics in Medicine*, *21*(4), 589–624. https://doi.org/10.1002/sim.1040

van Houwelingen, H. C., Zwinderman, K. H., & Stijnen, T. (1993). A bivariate approach to meta-analysis. *Statistics in Medicine*, *12*(24), 2273–2284. https://doi.org/10.1002/sim.4780122405

Viechtbauer, W. (2007). Confidence intervals for the amount of heterogeneity in meta-analysis. *Statistics in Medicine*, *26*(1), 37–52. https://doi.org/10.1002/sim.2514

Wang, L., Fang, Y., & Bergeman, C. S. (2026). Dynamic modeling with intensive longitudinal data: One-step and two-step DSEM approaches. *Structural Equation Modeling: A Multidisciplinary Journal*, 1–17. https://doi.org/10.1080/10705511.2025.2597284

Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology*, *54*(6), 1063–1070. https://doi.org/10.1037/0022-3514.54.6.1063

Yoo, A. B., Jette, M. A., & Grondona, M. (2003). SLURM: Simple Linux Utility for Resource Management. In D. Feitelson, L. Rudolph, & U. Schwiegelshohn (Eds.), *Job scheduling strategies for parallel processing* (pp. 44–60). Springer Berlin Heidelberg. https://doi.org/10.1007/10968987_3

You, D., Hunter, M., Chen, M., & Chow, S.-M. (2019). A diagnostic procedure for detecting outliers in linear state-space models. *Multivariate Behavioral Research*, *55*(2), 231–255. https://doi.org/10.1080/00273171.2019.1627659

Zhang, G., Lee, D., Li, Y., & Ong, A. (2025). Dynamic factor analysis with multivariate time series of multiple individuals: An error-corrected estimation method. *Psychological Methods*. https://doi.org/10.1037/met0000722

Figure 1

Relative Bias for the Fixed Effects

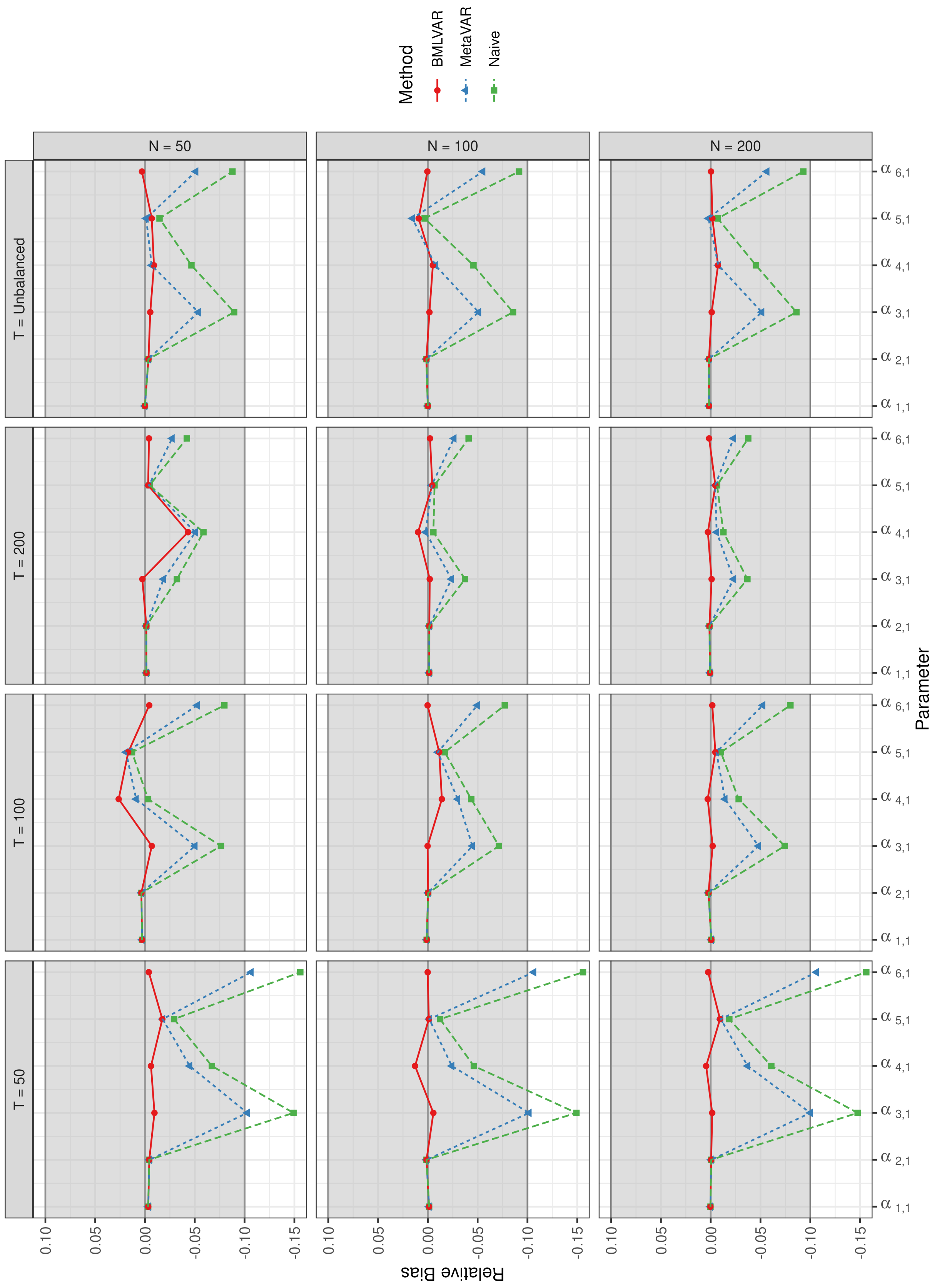

Figure 2

Relative Bias for the Random Effects

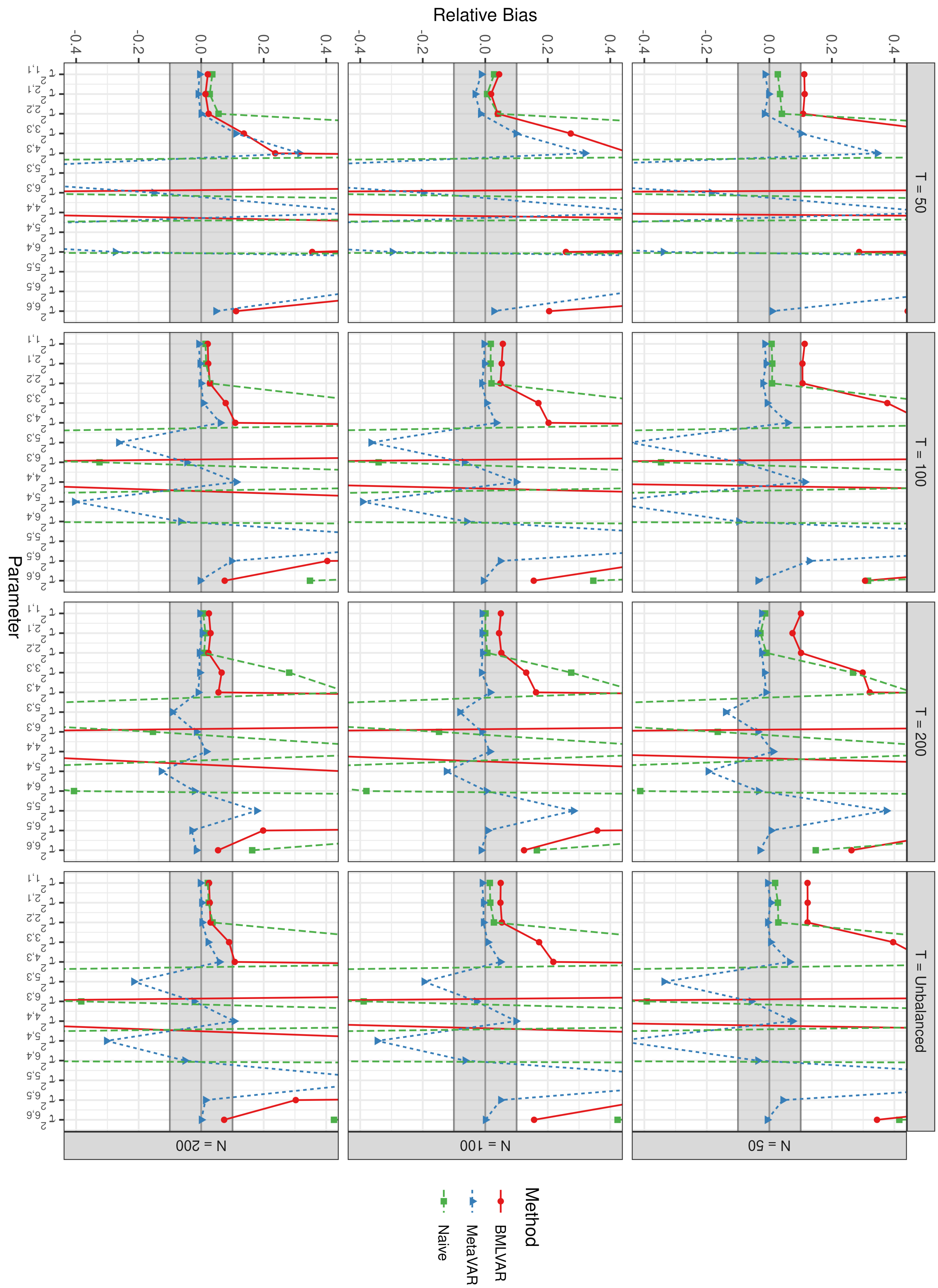

Figure 3

Root Mean Square Error for the Fixed Effects

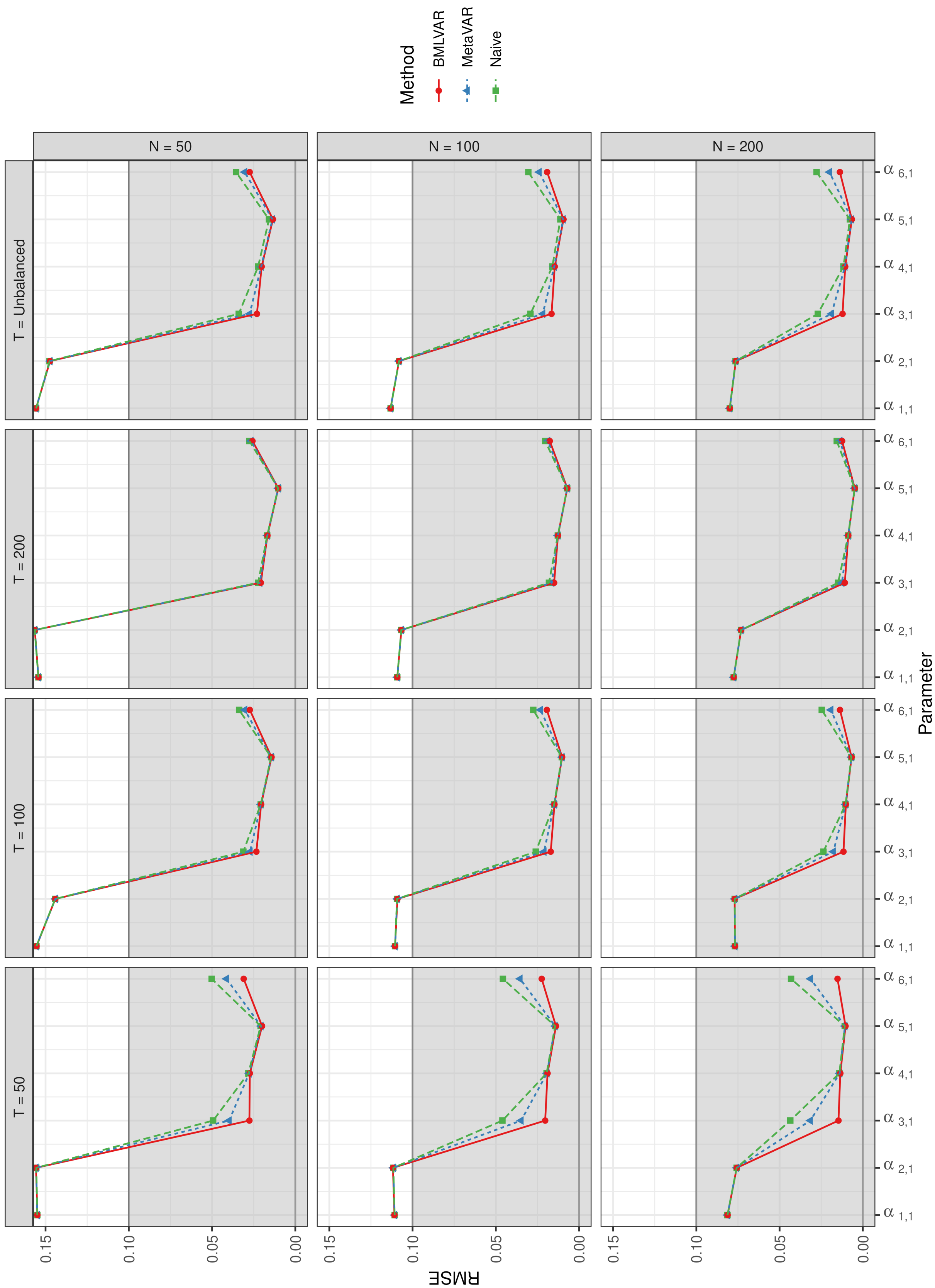

Figure 4

Root Mean Square Error for the Random Effects

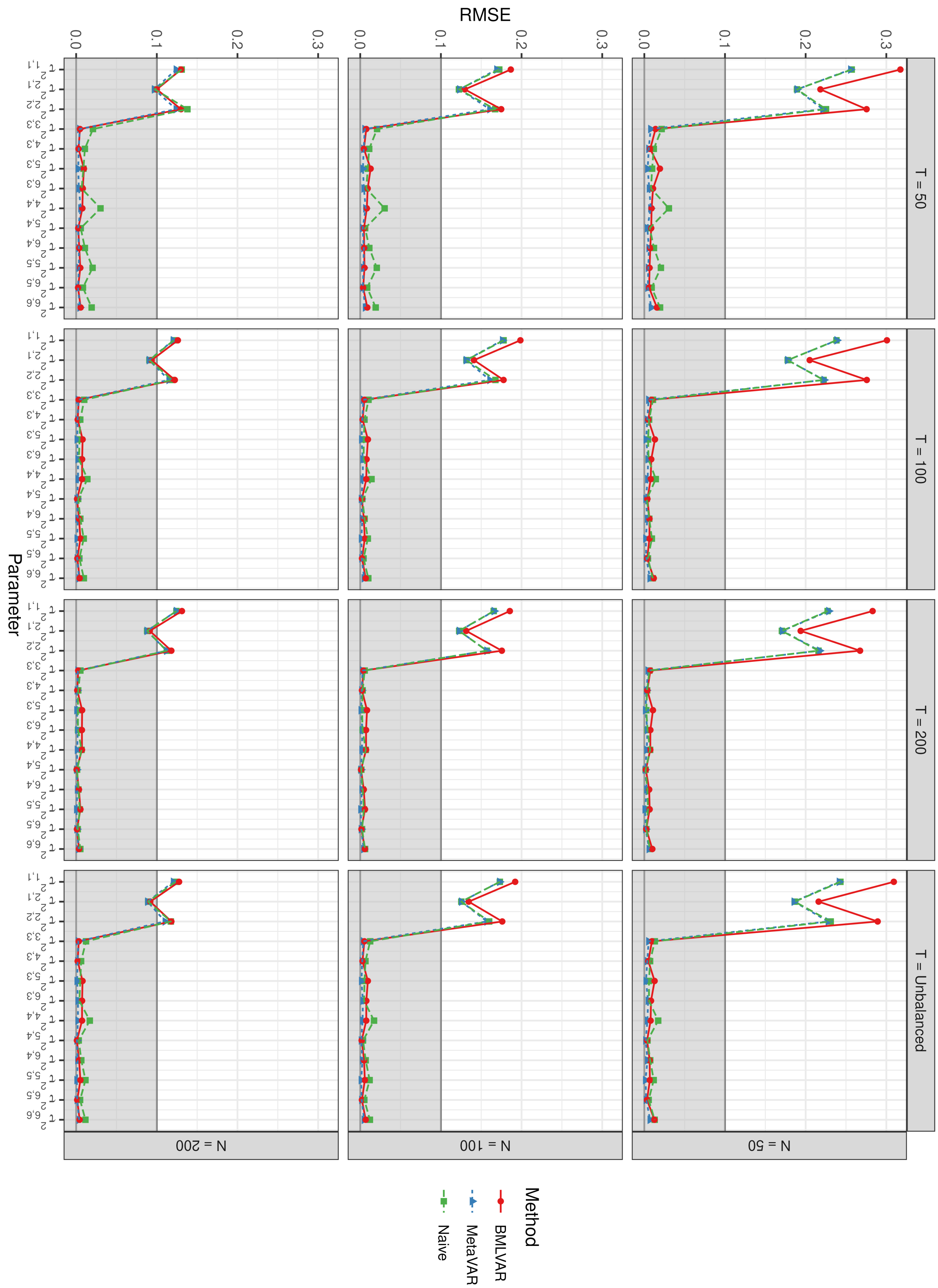

Figure 5

Coverage Probability for the Fixed Effects

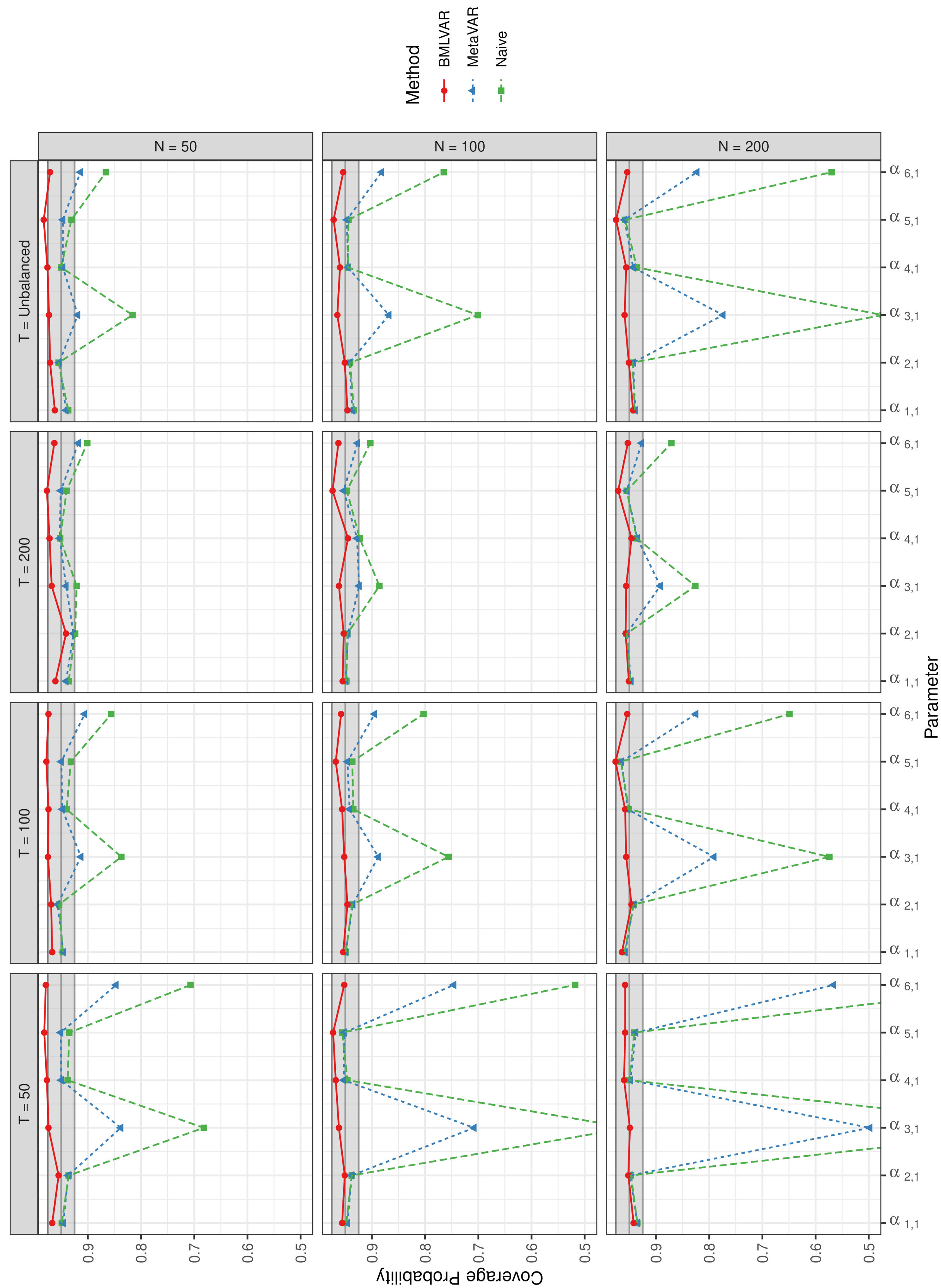

Figure 6

Coverage Probability for the Random Effects

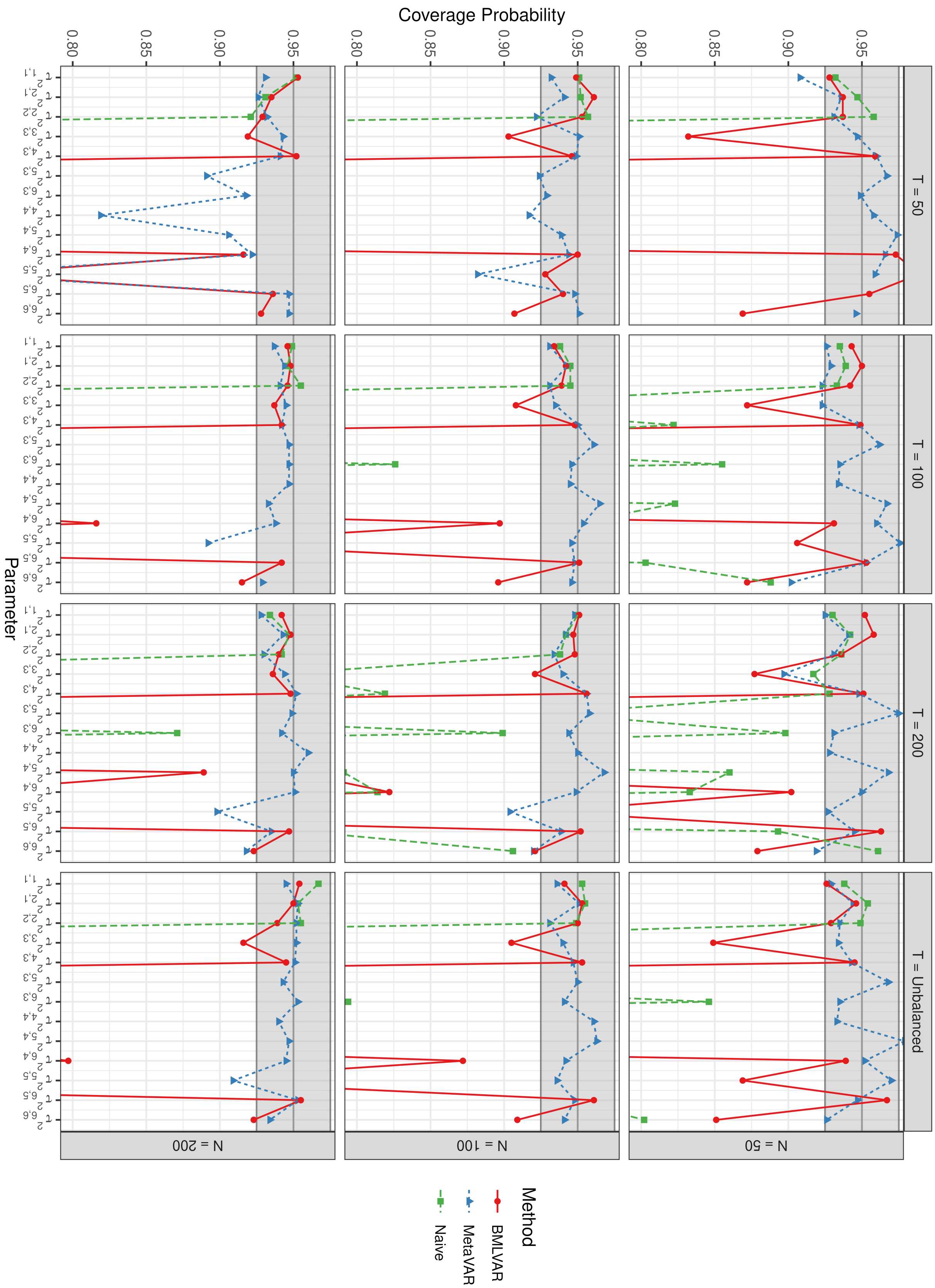

Figure 7

Statistical Power for the Fixed Effects

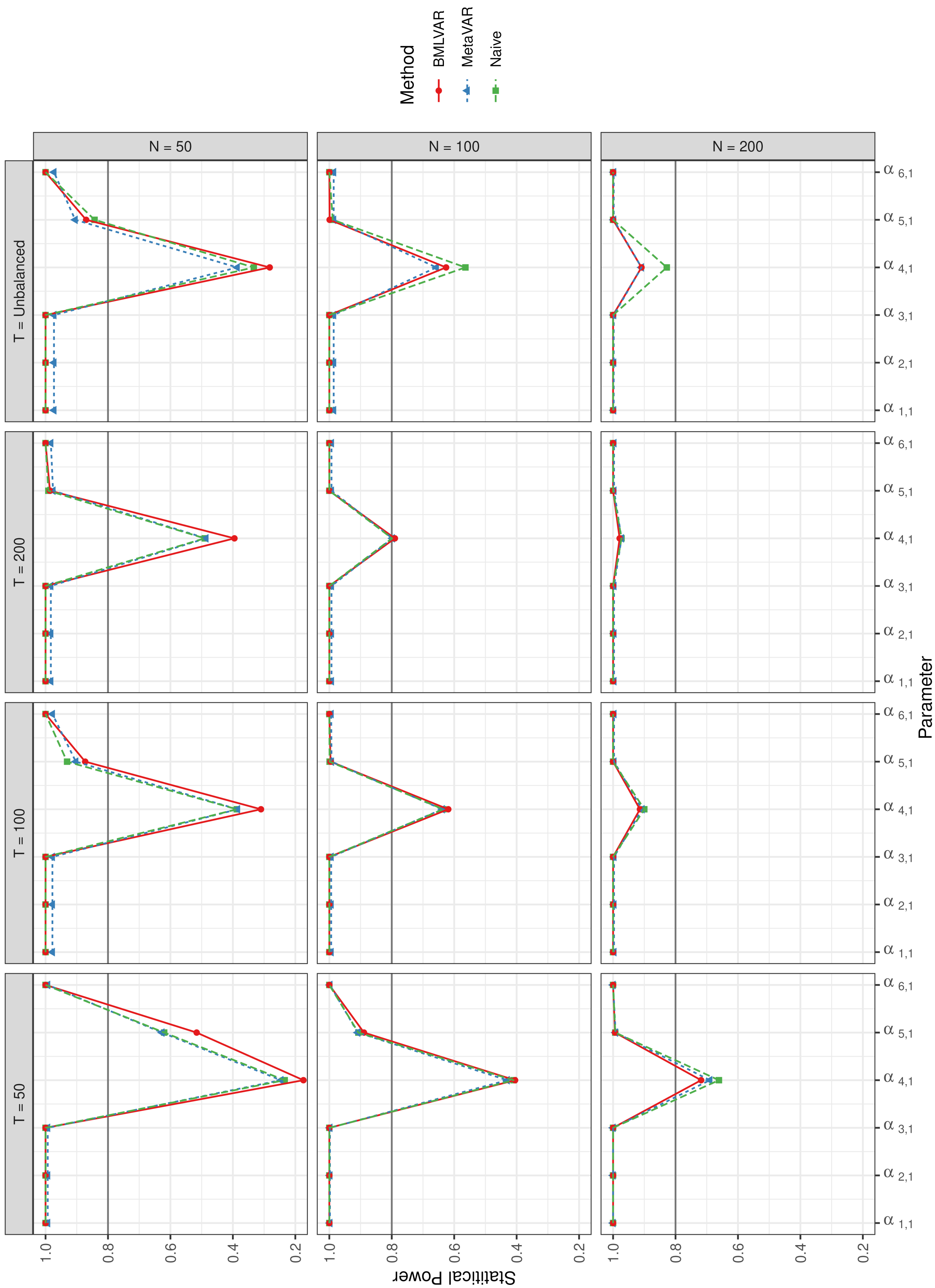

Figure 8

Statistical Power for the Random Effects